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MODULAR WAGON

Abstract

A wagon assembly includes a wagon chassis having a frame with a first end, a second, a left side, and a right side, a handle coupled to the frame, a deck coupled to the frame, the deck having a population of openings arranged in a plurality of rows and columns, and a first pair of wheels coupled to the frame. The wagon chassis may also include a second pair of wheels coupled to the frame. At least one opening of the population of openings in the deck is configured to receive a peg of an additional component to couple the additional component to the wagon chassis. The wagon chassis may include first and second chassis portions rotatably connected to one another, wherein the wagon chassis is configured to fold and unfold along a fold axis extending from the left side to the right side of the wagon chassis.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS [0001] This application claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application Ser. No. 63/559,789, filed on Feb. 29, 2024 and U.S. Provisional Patent Application Ser. No. 63/718,840, filed on Nov. 11, 2024, the entire contents of each of which are incorporated herein by reference. [0002] This application is related to commonly owned U.S. application Ser. No. 17/456,072, filed Nov. 22, 2021, U.S. application Ser. No. 16/686,454, filed Nov. 18, 2019, U.S. application Ser. No. 15/453,329, filed Mar. 8, 2017, U.S. Provisional Patent Application Ser. No. 63/718,840, filed on Nov. 11, 2024, U.S. Provisional Patent Application Ser. No. 63/559,789, filed Feb. 29, 2024, U.S. Provisional Patent Application Ser. No. 62/307,726, filed Mar. 14, 2016, and U.S. Provisional Patent Application Ser. No. 62/305,017, filed Mar. 8, 2016, the entire contents of each of which are incorporated herein by reference. [0003] This application is related to commonly owned U.S. application Ser. No. 14/444,380, filed Jul. 28, 2014, U.S. application Ser. No. 14/811,340, filed Jul. 28, 2015, and U.S. application Ser. No. 14/992,899 filed Jan. 11, 2016, the entire contents of each of which are incorporated herein by reference.

FIELD

[0004] The present invention relates generally to consumer-use wagons, and particularly to pull-handle and push handle wagons for multi-purpose use including carrying seated children and/or cargo.

BACKGROUND

[0005] The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

[0006] Conventional consumer-use wagons are commonly used for holding and carrying cargo such as children. Such wagons typically include a generally rectangular base and four generally rectangular upright walls forming an open-topped container, with a pull handle pivotally coupled to the base front, and with four wheels rotationally mounted to the base bottom. A traditional and well-known wagon of this type is the classic RADIO FLYER wagon.

[0007] While these wagons have their advantages, they also have some drawbacks. Accordingly, needs exist for improved features for consumer-use wagons for multi-purpose use. It is to the provision of solutions to these and other problems that the present invention is primarily directed.

SUMMARY

[0008] This section provides a general summary of the disclosure, and is not a comprehensive disclosure of its full scope or all of its features.

[0009] Generally described, the present invention relates to consumer-use wagons for multi-purpose use including carrying seated children and/or cargo, including but not limited to coolers, chairs, fishing equipment (e.g., rods or spears), skis, paddleboards, crates for pets, yard waste bins, smaller containers for storage or temperature control or the like, etc. In some embodiments, the wagons may be modular wagons that may comprise one or more modular wagon elements that together may be combined to define a wagon having a particular shape, a particular combination of containers, or other features. In some embodiments, the wagons may have a footwell for children carried in the wagon to place/rest their feet, and the footwell is repositionable between an extended/use position lowered relative to the wagon base and a collapsed/storage position closer to the wagon base. And in some embodiments, the wagons have a handle that can be moved between different positions and that is lockable in multiple of the different positions to allow the wagon to be easily pushed or pulled.

[0010] According to one exemplary embodiment, a wagon assembly includes a modular wagon bed including at least one sidewall positionable in a use position and a stored position. The modular wagon bed may include a footwell feature positionable in an extended position and a storage position. The wagon assembly includes a wagon chassis including a pivotable handle, a plurality of wheels, and a deck, wherein the modular wagon bed is removably coupled to the wagon chassis.

[0011] The wagon assembly may include various optional embodiments. The deck may include a movable surface for accommodating the footwell feature of the modular wagon bed when the modular wagon bed is coupled to the wagon chassis. The wagon assembly may include one or more adapters configured to removably couple to the wagon chassis and configured to removably couple to an accessory. The wagon assembly may further include one or more adapters configured to removably couple to the modular wagon bed and configured to removably couple to an accessory. The footwell feature may further include a footrest coupled to at least one footwell sidewall and, in a storage position, the footrest may be positioned more proximate to at least one seating surface of the wagon assembly than the footrest when the footwell feature is in an extended position. The handle may be configured to transition from a first use position and a stowed position, and the handle may include a pivotable coupling for pivoting the handle between the first use position and the stowed position. The handle may be positionable in a second use position via the pivotable coupling for pivoting the handle between the first use position and the second use position. The handle may be couplable to each endwall of the wagon chassis.

[0012] According to another exemplary embodiment, a wagon assembly includes a wagon chassis including a pivotable handle, a plurality of wheels, and a deck coupled to the plurality of wheels and the pivotable handle. The deck includes a footwell feature positionable in an extended position and a storage position.

[0013] The wagon assembly may include various optional embodiments. The wagon assembly may further include one or more adapters configured to removably couple to the wagon chassis and configured to removably couple to an accessory. The accessory may be a child seat. The accessory may be a cooler.

[0014] According to yet another exemplary embodiment, a wagon assembly includes a wagon chassis including a pivotable handle, a plurality of wheels, a center support coupled to the pivotable handle, and a deck. Each of the plurality of wheels is coupled to the center support.

[0015] The wagon assembly may include various optional embodiments. The wagon assembly may further include one or more adapters configured to removably couple to the deck and configured to removably couple to an accessory. The wagon assembly may further include at least two adapters configured to removably couple to the deck for mounting at least two child seats to the deck.

[0016] In some exemplary embodiments, a foldable wagon assembly is provided that includes a wagon chassis. The wagon chassis includes a first chassis portion, having a first frame portion and a first deck panel, a handle coupled to the first chassis portion, and a first pair of wheels coupled to the first chassis portion. The wagon chassis further includes a second chassis portion, having a second frame portion and a second deck panel, the second chassis portion connected to the first chassis portion by a first hinge and a second hinge, the first hinge and the second hinge configured to permit the first chassis portion and the second chassis portion to rotate with respect to one another between a first position and a second position. The wagon chassis further includes a second pair of wheels coupled to the second chassis portion.

[0017] According to yet another exemplary embodiment, a wagon assembly, includes a wagon chassis, the wagon chassis including a frame having a first end, a second end opposite the first end, a left side, and a right side; a handle coupled to the frame; a deck coupled to the frame, the deck having a population of openings; and a first pair of wheels coupled to the frame.

[0018] Further areas of applicability and advantages will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations, and are not intended to limit the scope of the present disclosure.

[0020] FIGS. 1A-1G show a wagon having a wagon chassis, in accordance with embodiments of the present disclosure;

[0021] FIGS. 1H-1J show the wagon with the wagon chassis and a modular wagon bed with a collapsible footwell feature, in accordance with embodiments of the present disclosure.

[0022] FIGS. 2A-2F show a wagon chassis having a modular wagon bed removed, in accordance with embodiments of the present disclosure.

[0023] FIGS. 3A-3C show a wagon chassis with a collapsible footwell feature, in accordance with embodiments of the present disclosure.

[0024] FIGS. 4A-4B show a wagon chassis with a collapsible footwell feature, in accordance with embodiments of the present disclosure.

[0025] FIGS. 5A-5D show an optional accessory, the optional accessory being one or more removable pads, that may be used with the wagon, in accordance with embodiments of the present disclosure.

[0026] FIGS. 6A-6C show an optional accessory, the optional accessory being a bag, that may be used with the wagon, in accordance with embodiments of the present disclosure.

[0027] FIGS. 7A-7C show exemplary optional additional components that may be used with the wagon chassis, including exemplary adapters for use with the wagon chassis to couple child seats to the wagon chassis, in accordance with embodiments of the present disclosure.

[0028] FIGS. 8A-8E show a wagon chassis with an alternative pivotable handle, in accordance with embodiments of the present disclosure.

[0029] FIG. 9 show a wagon chassis, in accordance with embodiments of the present disclosure.

[0030] FIGS. 10A-10C show a retractable cargo deck, in accordance with embodiments of the present disclosure.

[0031] FIGS. 11A-11B show an exemplary adapter, in accordance with embodiments of the present disclosure.

[0032] FIGS. 12A-12B show an exemplary adapter, in accordance with embodiments of the present disclosure.

[0033] FIGS. 13A-13B show an exemplary adapter, in accordance with embodiments of the present disclosure.

[0034] FIGS. 14A-14B show an exemplary adapter, in accordance with embodiments of the present disclosure.

[0035] FIGS. 15A-15D show an exemplary adapter, in accordance with embodiments of the present disclosure.

[0036] FIG. 16 shows wheels of the wagon chassis, in accordance with embodiments of the present disclosure.

[0037] FIGS. 17A-17B show wheels of the wagon chassis, in accordance with embodiments of the present disclosure.

[0038] FIGS. 18A-18D show a wagon chassis, in accordance with embodiments of the present disclosure.

[0039] FIGS. 19A-19C show an optional accessory, the optional accessory being one or more storage devices, that may be used with the wagon, in accordance with embodiments of the present disclosure.

[0040] FIGS. 20A-20C show a wagon chassis, in accordance with embodiments of the present

disclosure.

[0041] FIG. **21** shows a wagon chassis, in accordance with embodiments of the present disclosure.

[0042] FIGS. **22A-22C** show a wagon chassis, in accordance with embodiments of the present disclosure.

[0043] FIGS. **23A-23E** show an exemplary adapter, in accordance with embodiments of the present disclosure.

[0044] FIGS. **24A-24B** show an exemplary adapter, in accordance with embodiments of the present disclosure.

[0045] FIGS. **25A-25C** show an exemplary adapter, in accordance with embodiments of the present disclosure.

[0046] FIGS. **26A-26B** show an exemplary adapter, in accordance with embodiments of the present disclosure.

[0047] FIGS. **27A-27D** show exemplary pegs, in accordance with embodiments of the present disclosure.

[0048] FIG. **28** shows an optional accessory, the optional accessory being an open top bin, that may be used with the wagon, in accordance with embodiments of the present disclosure.

[0049] FIGS. **29A-29B** show the wagon chassis configured as a trailer, in accordance with embodiments of the present disclosure.

[0050] FIGS. **30A-30B** show the wagon chassis configured as a trailer, in accordance with embodiments of the present disclosure.

[0051] FIG. **31** shows the wagon chassis configured as a trailer, in accordance with embodiments of the present disclosure.

[0052] FIGS. **32A-32B** show the wagon chassis configured as a trailer, in accordance with embodiments of the present disclosure.

[0053] FIGS. **33A-33D** show exemplary locking hinges of the wagon chassis, in accordance with embodiments of the present disclosure.

[0054] FIGS. **34A-34B** show exemplary locking hinges of the wagon chassis, in accordance with embodiments of the present disclosure.

[0055] FIGS. **35A-35D** show exemplary locking hinges of the wagon chassis, in accordance with embodiments of the present disclosure.

[0056] FIGS. **36A-36C** show exemplary adjustable ports and exemplary accessories of the wagon chassis, in accordance with embodiments of the present disclosure.

[0057] Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

[0058] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.

[0059] Example embodiments are provided so that this disclosure will be thorough, and will fully convey the scope to those who are skilled in the art. Numerous specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed, that example embodiments may be embodied in many different forms and that neither should be construed to limit the scope of the disclosure. In some example embodiments, well-known processes, well-known device structures, and well-known technologies are not described in detail.

[0060] The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of stated features, integers, steps, operations, elements, and/or

components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. It is also to be understood that additional or alternative steps may be employed.

[0061] When an element or layer is referred to as being “on,” “engaged to,” “connected to,” or “coupled to” another element or layer, it may be directly on, engaged, connected or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0062] Although the terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections, these elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms when used herein do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example embodiments.

[0063] Spatially relative terms, such as “inner,” “outer,” “beneath,” “below,” “lower,” “above,” “upper,” “front,” “rear,” “left,” “right,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. Spatially relative terms may be intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the example term “below” can encompass both an orientation of above and below. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly.

Wagon

[0064] Generally described, the present invention relates to a consumer-use wagon that includes a base and a peripheral wall arrangement that together form a container, and that additionally includes a collapsible footwell, a position-locking handle, or both. It will be understood that the specific dimensions and proportions identified herein are representative for illustration purposes only and are not limiting of the invention; in fact, the invention expressly contemplates and covers many other embodiments with different dimensions, proportions, assembly methods, and overall designs. Additionally, although various embodiments of the wagon are described herein as “modular wagon,” “wagon,” or a “wagon assembly,” the embodiments may also be referred to as a “wheeled conveyance device.”

[0065] Various embodiments of the wagon or wagon chassis described herein may be modular such that one or more wagon elements or features may be assembled together to define a wagon having a particular shape or particular receptacle feature (e.g., for receiving one or more beach chairs, surfboards, paddle boards, paddles, fishing rods, coolers, yard waste bins, fishing equipment, skis, etc.). The modular wagon may be assembled to provide a shape and design (e.g., particular containers, features, etc.) designed to carry a variety of items that may connect either directly to the wagon chassis or indirectly (e.g., using an adapter) or which may be retained in place via walls or other features of the wagon (including walls or other features of a modular container couplable to

the wagon).

[0066] In some embodiments, the wagon chassis couples to a modular wagon. The modular wagon or the wagon chassis may include a collapsible footwell feature according to some embodiments. Embodiments of the present disclosure provide that the walls of the modular wagon and/or the collapsible footwell feature may be removed from the wagon chassis or the walls and/or footwell may remain on the wagon chassis and be folded down. The wagon chassis described herein may couple to a modular wagon including a footwell that may be removable from the modular wagon. In some embodiments, the wagon chassis may include a removable or slidable panel that can be closed or opened depending on whether the footwell is desired. Various adapters or modular elements may be removably coupled to the wagon chassis for mounting one or more child seats, benches, or stroller seats when the walls are folded or otherwise removed. At least some of the embodiments of the present disclosure provide for extending the length of the wagon, via expanding the length of the wagon chassis, for accommodating larger loads such as longer seats or larger coolers. Additional modular elements or accessory attachments (referred to interchangeably herein as adapters) may be provided for coupling directly or indirectly to, or may provide for a container for receiving, a variety of commonly carried items such as coolers, chairs, fishing equipment (e.g., rods or spears), skis, paddleboards, crates for pets, yard waste bins, smaller containers for storage or temperature control or the like, etc. At least some of the adapters may be configured to couple accessories to one another to increased space efficiency and securement. The accessories and adapters described herein are optional and the accessories and adapters may collectively be referred to as optional additional components. The optional additional components may be used according to the needs and desires of the user of the modular wagon **100**.

[0067] FIGS. **1A-1J** illustrate an exemplary embodiment of a wagon **100**. The wagon **100** includes a wagon chassis **104** and one or more optional accessories, such as for example a modular wagon bed **106** (see FIGS. **1H** and **1I**) or adapters (see FIGS. **7A-7C**, **11A-11B**, **12A-12B**, **13A-13B**, **14A-14B**, **15A-15D**), that can be removably coupled to the wagon chassis **104**. The wagon chassis **104** and the optional additional components describe herein may collectively be referred to as a “modular wagon,” “wagon,” or a “wagon assembly” according to various embodiments of the present disclosure. Additionally, it will be understood that the wagon chassis **104** alone may be referred to as a “modular wagon,” “wagon,” or a “wagon assembly” according various embodiments of the present disclosure. As shown herein, in many embodiments, the wagon chassis **104** is a foldable wagon chassis **100**. However, in other embodiments the modular wagon **100** is not foldable.

[0068] The wagon chassis **104** includes a frame **2716**, a deck **2720**, wheels **112a**, **112b**, and a handle **114**. The deck **2720** is coupled to the frame **2716** and forms a support surface for the optional accessories. In some embodiments, the deck **2720** may include one or more deck panels **2722** that are removably coupled to the frame **2716**. The wheels include a first pair of wheels **112a** coupled to the frame **2716** and a second pair of wheels **112b** coupled to the frame **2716**. In some embodiments, the first pair of wheels **112a** are coupled proximate to or at a first end of the frame **2716**, such as a front end of the frame **2716**, on opposite sides of the frame **2716**. In some embodiments, the first pair of wheels **112a** swivel with respect to the wagon chassis **104**. The first pair of wheels **112a** may be referred to as swivel wheels **112a**. In some embodiments, the second pair of wheels **112b** are coupled proximate to or at a second end of the frame **2716**, such as a rear end of the frame **2716**, on opposite side of the frame **2716**. In some embodiments, the second pair of wheels **112b** do not swivel with respect to the wagon chassis **104**. In some embodiments, the second pair of wheels **112b** have a larger diameter than the first pair of wheels **112a**.

[0069] The wheels **112a**, **112b** may be displaceable from their use position, for example by being removable from the frame **2716**, foldable flat to adjacent the deck **2720**, etc. It will be understood that different sizes of wheels **112a**, **112b** may be used at a front (e.g., proximate the handle **114**) compared to the back (e.g., opposite the handle **114**) for example, for providing increasing off road

capabilities and the ability for the wagon chassis **104** to be transported across varying terrains such as beaches, hiking trails, snow, or the like. Thinner or thicker wheels **112a**, **112b** may be used to further accommodate various terrains. Accordingly, any combination of size (e.g., diameter and/or width) of wheel **112a**, **112b** may be used with the wagon chassis **104** as described throughout the present disclosure.

[0070] The handle **114** is coupled to the frame **2716**. The handle **114** may pivot with respect to the frame **2716** and may fold down toward the deck **2720**. In some embodiments, the handle **114** is removably coupled to the frame **2716** and may alternatively be coupled to the first end of the frame **2716** and the second end of the frame **2716**.

Wagon Chassis

[0071] In some embodiments, the wagon chassis **104** has a first chassis portion **2708** and a second chassis portion **2712** that are configured to fold relative to one another. Coupled to the first chassis portion **2708** are the first pair of wheels **112a** and the handle **114**, and coupled to the second chassis portion **2712** are the second pair of wheels **112b**. Throughout this specification, the wagon chassis **104** may be referred to as having a front, a rear, a left side, and a right side; however, it should be understood that these terms are used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures, and that what may be considered the front, rear, left, and right of the wagon chassis **104** may be swapped depending on, for example, if the locations of the wheels **112a**, **112b** are swapped and/or if the location of the handle **114** is swapped, as described herein.

[0072] In some example embodiments, the wagon chassis **104** is configured to be selectively positioned in a first position or a second position. The first position is considered the unfolded or use position, and the second position is considered the folded or storage position. The first chassis portion **2708** and the second chassis portion **2712** are positionable between the first position and the second position, such that the first chassis portion **2708** and the second chassis portion **2712** are rotatable relative to one another between the first position and the second position. The foldable wagon chassis **104** is therefore able to be folded down to a smaller size for storage and or transport. In some embodiments, the frame **2716** of the wagon chassis **104** is not foldable.

Frame

[0073] In some embodiments, the frame **2716** of the wagon chassis **104** includes one or more frame members, one or more swivel wheel housings, and one or more hinges. For example, the frame **2716** includes a first frame member **2724**, a second frame member **2728**, a third frame member **2732**, a fourth frame member **2736**, a fifth frame member **2740**, and a sixth frame member **2744**. The first frame member **2724** may be considered a front frame member. The second frame member **2728** may be considered a rear frame member. The third frame member **2732** and fourth frame member **2736** may be considered left side frame members. The fifth frame member **2740** and sixth frame member **2744** may be considered right side frame members. The first frame member **2724**, third frame member **2732**, and fifth frame member **2740** cooperate to form a first frame portion **2748** of the frame **2716**. The second frame member **2728**, fourth frame member **2736**, and sixth frame member **2744** cooperate to form a second frame portion **2752**.

[0074] The frame **2716** also includes a first swivel wheel housing **2726**, a second swivel wheel housing **2730**, a third swivel wheel housing **2734**, and a fourth swivel wheel housing **2738**. The swivel wheel housings **2726**, **2730**, **2734**, and **2738** are located at the four corners of the wagon chassis **104**. The first frame member **2724** and the third frame member **2732** are coupled to the first swivel wheel housing **2726**, for example, by welds. The first frame member **2724** and the fifth frame member **2740** are coupled to the second swivel wheel housing **2730**, for example, by welds. The second frame member **2728** and the fourth frame member **2736** are coupled to the third swivel wheel housing **2734**, for example, by welds. The second frame member **2728** and the sixth frame member **2744** are coupled to the fourth swivel wheel housing **2738**, for example, by welds. Each of the swivel wheel housings **2726**, **2730**, **2734**, and **2738** have sockets that are configured to receive

the stems of the first pair of wheels **112a**. Although there are four swivel wheel housings **2726**, **2730**, **2734**, and **2738** shown, there are typically only two swivel wheel housings used at any given time. That is, the wagon chassis **104** usually only includes one pair of swivel wheels **112a**.

[0075] In some embodiments, the wagon chassis **104** further includes two hinges: a first hinge **2756**, and a second hinge **2760**. The first hinge **2756** couples the third frame member **2732** with the fourth frame member **2736**. The second hinge **2760** couples the fifth frame member **2740** with the sixth frame member **2744**. The first hinge **2756** may be considered a left side hinge and the second hinge **2760** may be considered a right side hinge. The first hinge **2756** and the second hinge **2760** allow relative rotation between the third frame member **2732** and the fourth frame member **2736** and the fifth frame member **2740** and the sixth frame member **2744**, respectively. Accordingly, the first hinge **2756** and the second hinge **2760** allow folding of the first frame portion **2748** and the second frame portion **2752** with respect to one another.

[0076] The first hinge **2756** and the second hinge **2760** are shown as being located on the wagon chassis **104** halfway between the front and rear of the wagon chassis **104**; however, in some embodiments, the first hinge **2756** and the second hinge **2760** may be located closer to the front of the wagon chassis **104** or closer to the rear of the wagon chassis **104** without departing from the scope of the disclosure.

[0077] As shown in FIGS. **1C** and **1D**, each of the first hinge **2756** and the second hinge **2760** include a first hinge element **2764** and a second hinge element **2768** that are pivotably coupled to one another by a hinge pin **2772**. The first hinge element **2764** includes, for example, a groove **2776** and the second hinge element **2768** includes a tab **2780**. The groove **2776** is sized and shaped to receive the tab **2780**. The first hinge element **2764** and the second hinge element **2768** rotate relative to one another about the hinge pin **2772**. For example, wagon chassis **104** is configured to fold and unfold along a fold axis extending from the left side to the right side of the wagon chassis **104**, the fold axis extending through the hinge pins **2772** of each of the first hinge **2756** and the second hinge **2760**.

[0078] In some embodiments, the first hinge **2756** and the second hinge **2760** each include a lock mechanism, such as a spring-loaded pin that is received in a corresponding hole, that prevents movement of the first hinge **2756** and the second hinge **2760** and holds the wagon chassis **104** in the first, unfolded, or use position. For example, the spring-loaded pin may be located in the groove **2776** of each first hinge element **2764** and may be received in a corresponding hole in the tab **2780** of each respective second hinge element **2768**. When a user desires to fold the wagon chassis **104**, the user can unlock the lock mechanism, using a fold actuator **2778** for example, to permit rotation of the first hinge **2756** and the second hinge **2760** and is then able to fold the wagon chassis **104** to the second, folded, or storage position. For example, the fold actuator **2778** may include a button that can be depressed to unlock the lock mechanism to fold the wagon chassis **104**. In some embodiments, the lock mechanism may also hold the wagon chassis **104** in the second, folded, or storage position. Thus, when a user desires to unfold the wagon chassis **104**, the user can unlock the lock mechanism to permit rotation of the first hinge **2756** and the second hinge **2760** and is then able to unfold the wagon chassis **104** to the first, unfolded, or use position.

Deck

[0079] With specific reference to FIGS. **1A** and **1B**, the deck **2720** includes one or more deck panels **2722** that cooperate to form the deck **2720**. The deck **2720** includes a top surface **2782** upon which items may be placed. For example, in some embodiments, the deck **2720** includes a first deck panel **2792**, a second deck panel **2796**, a third deck panel **2794**, and a fourth deck panel **2798**. The first deck panel **2792**, the second deck panel **2796**, the third deck panel **2794**, and the fourth deck panel **2798** each have a respective top surface and bottom surface, wherein the respective top surfaces of the first deck panel **2792**, the second deck panel **2796**, the third deck panel **2794**, and the fourth deck panel **2798** form the top surface **2782** of the deck **2720**. It should be understood that in embodiments, where the wagon chassis **104** includes a single deck panel, that the top surface of

that single deck panel forms the top surface **2782** of the deck **104**.

[0080] With continued reference to FIGS. **1A** and **1B**, in some embodiments, the first deck panel **2792** and the third deck panel **2794** are part of the first chassis portion **2708** and the second deck panel **2796** and the fourth deck panel **2798** are part of the second chassis portion **2712**. The first deck panel **2792** and the third deck panel **2794** are coupled to the first frame portion **2748** and the second deck panel **2796** and the fourth deck panel **2798** are coupled to the second frame portion **2752**. As an example, the first deck panel **2792** and the third deck panel **2794** may be coupled to and bounded by the first frame member **2724**, the third frame member **2732**, and the fifth frame member **2740**. The tops of the first deck panel **2792** and the third deck panel **2794** may be flush with the top of the first frame member **2724**, the third frame member **2732**, and the fifth frame member **2740**. Additionally, the second deck panel **2796** and the fourth deck panel **2798** may be coupled to and bounded by the second frame member **2728**, the fourth frame member **2736**, and the sixth frame member **2744**. The tops of the second deck panel **2796** and the fourth deck panel **2798** may be flush with the top of the second frame member **2728**, the fourth frame member **2736**, and the sixth frame member **2744**. When the wagon chassis **104** is in the first, unfolded, or use position, the top of the first deck panel **2792** is flush with the top of the second deck panel **2796**.

[0081] In some embodiments, the first deck panel **2792**, the second deck panel **2796**, the third deck panel **2794**, and the fourth deck panel **2798** are fixedly coupled to the frame **2716**. In some embodiments, the first deck panel **2792** and the second deck panel **2796** are fixedly coupled to the frame **2716**, while the third deck panel **2794** and the fourth deck panel **2798** are removably coupled to the frame **2716**. In some embodiments, one, some, or all of the first deck panel **2792**, the second deck panel **2796**, the third deck panel **2794**, and the fourth deck panel **2798** are fixedly coupled to the frame **2716**. In some embodiments, one, some, or all of the first deck panel **2792**, the second deck panel **2796**, the third deck panel **2794**, and the fourth deck panel **2798** are removably coupled to the frame **2716**. In some embodiments, the first deck panel **2792** and the second deck panel **2796** are fixedly in position to the frame **2716**, while the third deck panel **2794** and the fourth deck panel **2798** are moveable with respect to the frame **2716**.

[0082] For example, in some embodiments, the deck **2720** is coupled to the wagon chassis **104** by fasteners, such as for example, screws, bolts, etc. that secure the one or more deck panels **2722** to the frame **2716**. In some embodiments, one or more tools may be required to remove and reattach the fasteners to decouple and couple the one or more deck panels **2722** to the wagon chassis. In some embodiments, the one or more deck panels **2722** include spring-loaded ball locks that are configured to be received in one or more corresponding detents in the frame **2716** of the wagon chassis **104** (an example of a ball lock is described herein with respect to FIG. **27B**). In some embodiments, the one or more deck panels **2722** are clamped to the frame **2716**. In some embodiments, the frame **2716** includes one or more ledges (see, e.g., ledge **3518** in FIG. **35A**) around the interior perimeter of the frame **2716** on which the one or more deck panels **2722** may be supported and/or coupled to. In some embodiments, the frame **2716** includes one or more crossmembers. For example, fasteners may extend through the top surface **2782** of the deck **2720** and into corresponding threaded holes in the one or more ledges and/or the one or more crossmembers of the frame **2716**. As another example, in some embodiments, the one or more deck panels **2722** may include one or more magnets which may be magnetically attracted to the frame **2716**, the one or more ledges, the one or more crossmembers, and/or corresponding magnets or magnetic material (e.g., ferrous metal, a plastic magnet) in or on the frame **2716**, the one or more ledges, and/or the one or more crossmembers. As such, the deck **2720** may be magnetically coupled to the wagon chassis **104**.

[0083] In some embodiments the deck **2720** is non-slip. For example, the deck **2720** may be made of a rubberized non-slip material and/or coated with a rubberized non-slip material. In some embodiments, the deck **2720** may have non-slip portions, such as for example, areas of high surface roughness. In some embodiments, the deck **2720** may include grooves or a tread pattern that

provide a non-slip surface.

Attachment Features

[0084] In the exemplary embodiment shown in FIGS. 1A and 1B, the deck 2720 includes a population of attachment features 2802 that are configured to permit the optional additional components, such as accessories and/or adapters, to be attached to the wagon chassis 104. For example, each of the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798 include a population of attachment features 2802 that are configured to permit accessories to be attached to the wagon chassis 104. For example, the population of attachment features 2802 includes, without limitation, one or more of: a population of tie downs 2804, and a population of openings 2808. The population of openings 2808 may be referred to as a smart grid.

[0085] The tie downs 2804 include two openings separated by a bar 2806. Alternatively, the tie downs 2804 may be considered a single opening bisected by a bar 2806. The tie downs 2804 permit, for example, bungee cords and/or cargo nets to be secured to the wagon chassis 104. The tie downs 2804 are shown proximate to the peripheral edges of the first deck panel 2792 and the second deck panel 2796. As shown, the tie downs 2804 are open from the top sides to the bottom sides of the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798.

[0086] Each of the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798 are shown as including two tie downs 2804. The tie downs 2804 in each of the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798 are located near the peripheries of the respective first deck panel 2792, second deck panel 2796, third deck panel 2794, and fourth deck panel 2798 proximate to the frame 2716.

[0087] Although, the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798 are shown as each having two tie downs 2804, it will be understood that each of the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798 may include less than two or greater than two tie downs 2804 without departing from the scope of the disclosure. Additionally, in some embodiments, the number of tie downs 2804 in the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798 may be different. In some embodiments, tie downs 2804 are only included in the first deck panel 2792, while in other embodiments, tie downs 2804 are only included in the second deck panel 2796, while in yet other embodiments, tie downs 2804 are only included in the first deck panel 2792 and the second deck panel 2796.

[0088] With continued reference to FIGS. 1A and 1B, the deck 2720 includes a population of openings 2808, which may be referred to as a smart grid. For example, the openings 2808 are shown as arrayed in a grid pattern through the one or more deck panels 2722 of the deck 2720 (e.g., the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798). As shown, the openings 2808 are open from the top sides to the bottom sides of the one or more deck panels 2722 of the deck 2720 (e.g., the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798). The openings 2808 are shown as being located in the central area of each of the one or more deck panels 2722 of the deck 2720 (e.g., the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798). In some embodiments, the openings 2808 are in an array lengthwise and widthwise on the one or more deck panels 2722 of the deck 2720 (e.g., the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798). Stated another way, the openings 2808 may be arranged in a plurality of rows and columns extending at least across a portion of the length and width of one or more deck panels 2722 of the deck 2720 (e.g., the first deck panel 2792, the second deck panel 2796, the third deck panel 2794, and the fourth deck panel 2798). In some embodiments, openings 2808 in adjacent rows may be

aligned or offset. For example, adjacent rows may be offset, as illustrated in FIGS. 1A and 1B, so that the openings **2808** are aligned in alternating rows. As described in greater detail herein, for example, pegs of the additional components are sized, shaped, and/or located to insert into one or more of the openings **2808** to couple the additional components to the wagon chassis **104** and the wagon **100** more generally. In some embodiments, the openings **2808** are arranged in non-linear patterns to receive pegs of specific additional components that are unique to that additional component. In some embodiments, the openings **2808** may be sized and/or shaped to permit rope to be extended through and/or tied to the openings **2808**. Although the openings **2808** are shown as hexagonal holes, it will be understood that the openings **2808** may have a different shape (e.g., circular, square, triangular, octagonal, ovular, stadium, trapezoidal) without departing from the scope of the disclosure.

[0089] Although, the first deck panel **2792**, the second deck panel **2796**, the third deck panel **2794**, and the fourth deck panel **2798** are shown as each having a population of openings **2808**, it will be understood that the number of openings **2808** in the first deck panel **2792**, the second deck panel **2796**, the third deck panel **2794**, and the fourth deck panel **2798** may be different. In some embodiments, openings **2808** are only included in the first deck panel **2792**, while in other embodiments, openings **2808** are only included in the second deck panel **2796**, while in yet other embodiments, openings **2808** are only included in the first deck panel **2792** and the second deck panel **2796**.

[0090] In addition to serving as attachment features **2802**, one or more of the tie downs **2804** and openings **2808** may provide ventilation and/or drainage for the wagon chassis **104**.

Foldable Wagon Chassis

[0091] From the foregoing description, it will be understood that in some embodiments, the first chassis portion **2708** of the wagon chassis **104** comprises the first frame portion **2748**, the first deck panel **2792**, two wheels **112a**, and the handle **114**, and the second chassis portion **2712** of the wagon chassis **104** comprises the second frame portion **2752**, the second deck panel **2796**, and two additional wheels **112b**. Additionally, it will be understood that, in some embodiments, the wagon chassis **104** includes four tie downs **2804** along the left side of the wagon chassis **104** and four tie downs **2804** along the right side of the wagon chassis **104**. In some embodiments, the wagon chassis **104** includes two tie downs **2804** along the front of the wagon chassis **104** and two tie downs **2804** along the rear of the wagon chassis **104**. Moreover, it will be understood that, in some embodiments, the wagon chassis **104** includes a population of openings **2808** in the deck **2720** of the wagon chassis **104**. Additionally, although the wagon chassis **104** is described as having a separate first frame portion **2748** and a separate first deck panel **2792** and a separate second frame portion **2752** and a separate second deck panel **2796**, it will be understood that in some embodiments the first frame portion **2748** and the first deck panel **2792** may be integrally formed with one another and the second frame portion **2752** and the second deck panel **2796** may be integrally formed with one another.

[0092] As shown in FIGS. 1A, when the wagon chassis **104** is in the first, unfolded, or use position, the top of the first deck panel **2792**, the top of the second deck panel **2796**, the top of the third deck panel **2794**, and the top of the fourth deck panel **2798** are parallel and flush with one another. Additionally, when the wagon chassis **104** is in the first, unfolded, or use position, the tops of the first deck panel **2792** and the third deck panel **2794** and the tops of the second deck panel **2796** and the fourth deck panel **2798** may have an angle of 180 degrees between them.

[0093] As shown in FIG. 1G, when the wagon chassis **104** is in the second, folded, or storage position, the tops of the first deck panel **2792** and the third deck panel **2794** and the tops of the second deck panel **2796** and the fourth deck panel **2798** face away from one another and the bottoms of the first deck panel **2792** and the third deck panel **2794** and the bottoms of the second deck panel **2796** and the fourth deck panel **2798** are proximate to and face one another.

Additionally, in some embodiments, when the wagon chassis **104** is in the second, folded, or

storage position, the tops of the first deck panel **2792** and the third deck panel **2794** and the tops of the second deck panel **2796** and the fourth deck panel **2798** may have an angle from 360 degrees to 300 degrees between them.

Handle

[0094] Referring now to FIGS. **1E**, **1F**, and **1G**, in some embodiments, the handle **114** is telescopic. For example, the handle **114** includes a first handle portion **114a** and a second handle portion **114b**, wherein the first handle portion **114a** is configured to be received in the second handle portion **114b** and is configured to slide at least partially into and out of the second handle portion **114b**. The handle **114** further includes a handle bar **114c** at the top of the first handle portion **114a**, for example, the handle bar **114c** and the first handle portion **114a** forming a T-shape. The handle **114** further includes a clamp, such as for example, a quick release clamp **114d** coupled to the second handle portion **114b** that can be released to extend or retract the first handle portion **114a** out of or into the second handle portion **114b**. The handle **114** is shown in an extended position in FIG. **1E** and a retracted or collapsed position in FIG. **1F**. As shown in FIG. **1G**, the first handle portion **114a** can retract into the second handle portion **114b** so that the handle **114** does not extend far beyond the top of the wagon chassis **104** when it is folded. For example, when in the fully collapsed position, the handle bar **114c** may be located sufficient distance above the top of the wagon chassis **104** when the wagon chassis **104** is folded to permit a user to grab the handle bar **114c** to move or lift the wagon chassis **104**. In some embodiments, for example, the handle **114** may have a telescoping structure that permits it to collapse to at or vertically below the first chassis portion **2708** and the second chassis portion **2712**.

[0095] According to various embodiments, the handle **114** may be removably couplable to the wagon chassis **104** for placement at either end of the wagon chassis **104**. For example, the handle **114** may be placed on an end proximate to relatively larger wheels **112** for making it easier for a user to push the wagon chassis **104** or the handle **114** may be placed on an end proximate to relatively smaller wheels **112** for making it easier for a user to pull the wagon chassis **104**. The handle **114** may be a position-locking handle feature. On a typical wagon the handle is free to rotate about its axis until its movement is limited by the wagon geometry or the ground. But on the wagon chassis **104** of the present invention, the handle **114** locks in one or more positions. Locking handle design options include automatic lock or only when intentioned by the user. Any embodiments of the position-locking handle **114** may be used as described at least in U.S. Pat. No. 11,198,460 which is incorporated by reference herein.

Wagon with Wagon Chassis and Modular Wagon

[0096] FIGS. **1H-1J** show a wagon **100**, which may be a modular wagon, with a collapsible footwell feature **102**. The wagon **100** may include the wagon chassis **104** and an accessory, such as a modular wagon bed **106**, which may be removably coupled to the wagon chassis **104**, which collectively may be referred to as a “modular wagon,” “wagon,” or “wagon assembly” according to various embodiments of the present disclosure. The collapsible footwell feature **102** may be integrated with either the removable modular wagon bed **106**, as shown in FIG. **1H**, or with the wagon chassis **104** as shown and described with respect to other figures. The modular wagon bed **106** may include a base **108** and peripheral walls **110** extending upright from the base **108**. For example, the base **108** may include a peripheral frame and two opposing sidewalls and two opposing endwalls, with the sidewalls and endwalls all pivotably mounted at lower portions thereof to the base **108** from an upright or use position (as shown in FIG. **1H**) to their collapsed or storage position (as shown in FIG. **1J**) such that the walls are folded down adjacent (i.e., against or closer to) the base **108**. The collapsible footwell feature **102** may include a footrest **116** and upright elements **118** extending between the base **108** of the modular wagon bed **106** and the footrest **116**. The footwell feature **102** may include an opening in the base **108** (e.g., where there is no bottom wall mounted to the base **108**), which can have the same shape (e.g., rectangular) and size as the footrest **116** and be aligned directly above the footrest **116** (e.g., generally centrally positioned in

the modular wagon bed **106**). In at least some embodiments, the collapsible footwell feature **102** may include at least one footwell sidewall **120**. The footwell sidewall(s) **120** may, in the storage position, may be positioned more proximate to the at least one seating surface (e.g., the portions of the base **108** that are not the collapsible footwell feature **102**) than the footrest **116** when the collapsible footwell feature **102** is in the extended position.

[0097] As shown in FIG. **1H**, the third deck panel **2794** and the fourth deck panel **2798** may be removed or stowed to provide an opening **124** in the deck **2720** of the wagon chassis **104** through which the collapsible footwell feature **102** may pass downward through.

[0098] In various embodiments, the upright elements **118** are provided by at least two upright structural members, one forward and one rearward, which are each pivotably coupled at upper portions to the base **108** and pivotably coupled at lower portions to the footrest **116**. The pivotable couplings can be of a conventional design and assembly. The at least one forward upright structural member can be provided by two tubes, wires, rods, etc. at the two respective forward corners of the footrest, by a plate, panel, lattice, etc. extending laterally across the footrest, or by other conventional structural members, and the same applies to the at least one rearward upright structural member. In this way, the footwell can be collapsed from the extended/use position to the collapsed/storage position by swinging/pivoting the footrest **116** forward or rearward, respectively. Also, a footwell peripheral sidewall **120** of soft goods (e.g., fabric) can be provided surrounding the footrest **116** and defining the footwell interior.

[0099] The collapsible footwell includes a footrest that is positionable between an extended/use position lowered relative to the base and a collapsed/storage position closer to the base to provide a compact profile/arrangement to reduce the occupied footwell volume for storage and/or transport purposes. In the collapsed/storage position, the vacated space where the footwell was in the extended/use position effectively reduces the height dimension of the wagon with the wheels displaced (e.g., removed or folded to adjacent the base) to provide a smaller size for transportation and storage, for example during shipping from the manufacturer and/or the retailer, during storage in a user's garage, and/or during transport in a user's personal vehicle. The overall height of the wagon, with the wheels displaced and the upright walls folded down, is thus significantly reduced, thereby significantly reducing its package size for shipment from the manufacturer and/or retailer, and thereby significantly reducing shipping costs.

[0100] In other embodiments not depicted herein but contemplated by the invention, other conventional linkage systems can be implemented extending between and interconnecting the wagon base and the footrest to permit the footrest to compress, fold, slide, screw, or otherwise collapse from its extended/use position to its/storage position closer to the wagon base.

[0101] As further shown in FIG. **1H**, in some embodiments, the modular wagon bed **106** includes one or more pegs **122** extending downward from the base **108** away from the peripheral walls **110**. For example, the modular wagon bed **106** includes four pegs **122** at the corners of the base **108**. The pegs **122** are configured to insert into openings **2808** in the deck **2720** of the wagon chassis **104** to secure the modular wagon bed **106** to the wagon chassis **104**. The pegs **122** thus may be sized to fit within the openings **2808** in the deck **2720**. In some embodiments, the pegs **122** are sized to fit within the openings **2808** in the deck **2720** and have a shape corresponding to the shape of the openings **2808**. For example, where the openings **2808** are hexagonal, the pegs **122** are hexagonal and are sized to fit within the openings **2808**. In some embodiments, the pegs **122** are sized to fit within the openings **2808** in the deck **2720**, have a shape corresponding to the shape of the openings **2808**, and are located on the accessory, such as the modular wagon bed **106**, to permit inserting the pegs **122** into the openings **2808**. Providing pegs **122** that have the same shape as the openings **2808** may provide for a more secure connection of the modular wagon bed **106** onto the wagon chassis **104** and may reduce or eliminate movement of the modular wagon bed **106** with respect to the wagon chassis **104**. Additionally, providing two or more pegs **122** on the additional components, such as the modular wagon bed **106**, can limit or prevent rotational movement of the

additional components with respect to the wagon chassis **104**.

[0102] The handle **114** may be configured to transition from a first use position and a stowed position according to at least some embodiment. The handle **114** may include a pivotable coupling for pivoting the handle between the first use position and the stowed position. Further, the handle **114** may be positionable in a second use position via the pivotable coupling for pivoting the handle between the first use position and the second use position. The handle **114** may be coupled to any of the frame members **2724**, **2728**, **2732**, **2736**, **2740**, **2744** or any of the ends of the wagon chassis **104** described herein, to be illustrated in further detail with respect to other figures.

[0103] According to various embodiments, the portions of the base **108** (interchangeably referred to herein as a bottom surface) of the modular wagon bed **106** may be a seating surface. For example, a child may sit on the base **108** of the modular wagon that is not part of the collapsible footwell feature **102** and rest their feet within the collapsible footwell feature **102**. Similarly, a bottom surface of the wagon chassis **104** (e.g., the deck **2720** described herein) may be a seating surface when a collapsible footwell feature **102** is in an extended position.

[0104] FIG. **1H** illustrates a disassembled wagon **100** including the modular wagon bed **106** removed from the wagon chassis **104**. According to various embodiments, it would be advantageous to be able to remove the modular wagon bed **106** and use the wagon chassis **104** to transport large or uniquely shaped objects that may not otherwise be securely transported by a modular wagon bed **106** having peripheral walls as shown in FIG. **1H**. For example, a user may wish to transport coolers, chairs, fishing equipment such as rods or spears, skis, paddleboards, pets, yard waste, smaller containers, or the like. Accordingly, embodiments of the present disclosure provide a modular wagon **100** that is designed to carry a variety of items, such as those listed above, either directly to the wagon chassis **104** or indirectly via an adapter or the like, to be described in further detail below.

[0105] FIG. **1I** illustrates an assembled wagon **100** having the modular wagon bed **106** removably coupled to the wagon chassis **104** and in a storage position. For example, the modular wagon bed **106** may have the upright walls **110** collapsed into a storage position and the handle **114** may be further folded from an upright or use position to the storage position as shown in FIG. **1J**.

[0106] Additionally, as shown in FIG. **1J**, the modular wagon bed **106** may be foldable. For example, the two opposing sidewalls are configured to be folded to enable the modular wagon bed **106** to be folded over to and onto the deck **2720**. Additionally, the handle **114** can be folded down toward the deck **2720** to sandwich the modular wagon bed **106** against the deck **2720**.

Moveable and Removable Deck Panels

[0107] FIGS. **2A-2F** shows the wagon chassis **104** with the modular wagon bed **106** removed. As shown and described with respect to FIGS. **1A-1J** and **2A**, embodiments of the wagon chassis **104** include the deck **2720** with first deck panel **2792**, second deck panel **2796**, third deck panel **2794**, and fourth deck panel **2798**. In FIGS. **2B** and **2C**, the third deck panel **2794** and the fourth deck panel **2798** may be slid away from the center of the wagon chassis **104**, folded over or under and adjacent to the first deck panel **2792** and the second deck panel **2796**, or otherwise removed from the wagon chassis **104** when an additional component, such as modular wagon bed **106**, is coupled to the wagon chassis **104** and a collapsible footwell feature, such as collapsible footwell feature **102**, is in a use position as shown in FIGS. **1H** and **1I**. Alternatively, the third deck panel **2794** and the fourth deck panel **2798** may remain in an extended position, as shown in FIG. **2A**, when the modular wagon bed **106** is coupled to the wagon chassis **104** and the collapsible footwell feature **102** remains in a storage position (e.g., up against the modular wagon such that the footrest **116** is in the same plane as a base **108** of the modular wagon bed **106**).

[0108] FIG. **2B** illustrates an alternative view of the wagon chassis **104** of FIG. **2A**. FIG. **2B** illustrates one embodiment of the third deck panel **2794** and the fourth deck panel **2798** where the third deck panel **2794** and the fourth deck panel **2798** may be at least partially or fully slid away from the center of the wagon chassis **104** to form an opening **124** in the deck **2720** to make room

for, for example, the collapsible footwell feature **102** of the modular wagon bed **106**. The third deck panel **2794** and the fourth deck panel **2798** may be configured to be positioned at any position between the completely closed position shown in FIG. 2A and the completely open position shown in FIG. 2C. For example, a user may only need a small opening **124** in the deck **2720** for a smaller container or adapter to be inserted therein or a user may want to completely open the deck **2720** to provide room for an extended collapsible footwell feature **102**.

[0109] In some embodiments, as shown in FIGS. 2D-2F, the wagon chassis **104** includes only three deck panels: the first deck panel **2792**, the second deck panel **2796**, and the third deck panel **2794** wherein the third deck panel **2794** is large enough in size to span between the first deck panel **2792** and the second deck panel **2796**. That is, the third deck panel **2794** may be larger than the third deck panel **2794** shown in FIG. 2C. As shown in FIGS. 2E and 2F, the third deck panel **2794** may be at least partially or fully slid away from the center of the wagon chassis **104** to form an opening **124** in the deck **2720** to make room for, for example, the collapsible footwell feature **102** of the modular wagon bed **106**. The third deck panel **2794** may be configured to be positioned at any position between the completely closed position shown in FIG. 2D, the partially open position shown in FIG. 2E, and the completely open position shown in FIG. 2F. For example, a user may only need a small opening **124** in the deck **2720** for a smaller container or adapter to be inserted therein or a user may want to completely open the deck **2720** to provide room for an extended collapsible footwell feature **102**.

Collapsible Footwell

[0110] FIGS. 3A-3C show embodiments of the wagon chassis **104** in which the collapsible footwell **102** is part of the wagon chassis **104**. In some embodiments, the wagon chassis **104** includes one or more support structures **126** that extend down below the frame **2716** that are then able to support one or more of the deck panels (e.g., the third deck panel **2794** and the fourth deck panel **2798**) to form the bottom of the collapsible footwell upon which a person riding on the wagon chassis **104** can place their feet. For example, as shown in FIGS. 3A and 3B, the support structures **126** include a pair of U-shaped support bars **128** that are rotatably coupled to, for example, the third frame member **2732** the fourth frame member **2736** (which may be considered the left side frame members) and the fifth frame member **2740** and the sixth frame member **2744** (which may be considered right side frame members). Each of the support bars **128** includes a pair of legs **130** with a support crossbar **132** extending connected to and extending between the pair of legs **130**. The support bars **128** can be located in a first, stowed position in which the legs **130** of support bars **128** are substantially parallel or parallel to the deck **2720** and in a second, deployed position in which the legs **130** of the support bars **128** are at an angle **134** with respect to the deck **2720** and the support crossbar **132** is located a distance below the deck **2720**.

[0111] In the second, deployed position, the angle **134** of the legs **130** of the support bars **128** may be, for example, from about 90 degrees to about 110 degrees. As shown in FIG. 3B, the support bars **128** are shown being rotated downward from the first, stowed position to the second, deployed position. FIGS. 3A and 3C show the support bars **128** in the second, deployed position with the third deck panel **2794** and the fourth deck panel **2798** placed onto and supported by the support crossbars **132**. As shown in FIG. 3A, in some embodiments, the third deck panel **2794** and the fourth deck panel **2798** may be rotated 90 degrees with respect and oriented so that the extend fore and aft with respect to the wagon chassis **104** and then placed onto the support bars **128**. In this orientation, the third deck panel **2794** and the fourth deck panel **2798** can span from one support crossbar **132** to the other support crossbar **132** without falling down. In some embodiments, the third deck panel **2794** and the fourth deck panel **2798** may be releasably coupled to the support bars **128**, for example, by a snap-fit connection between the third deck panel **2794** and the fourth deck panel **2798** and the support crossbars **132**. As described herein, it will be understood that in some embodiments, the third deck panel **2794** and the fourth deck panel **2798** may be substituted with a single the third deck panel that is the same size and shape as the third deck panel **2794** and the

fourth deck panel **2798**.

[0112] Referring now to FIGS. **4A** and **4B**, another embodiment of support structures **126** are shown. For example, as shown in FIGS. **4A** and **4B**, the support structures **126** include a pair of straps **136** that are connected to the wagon chassis **104**. In some embodiments, the pair of straps **136** that are connected to the frame **2716**. In some embodiments, pair of straps **136** that are connected to the underside of the first deck panel **2792** and the second deck panel **2796**. In this embodiment, the third deck panel **2794** and the fourth deck panel **2798** are supported by the pair of straps **136**. The third deck panel **2794** and the fourth deck panel **2798** may be fixedly or releasably coupled to the straps **136**, for example, by releasable snap buttons, slots in the third deck panel **2794** and the fourth deck panel **2798** through which the straps **136** extend, rivets, screws, bolts, etc. The pair of straps **136** can be loosened to increase their length, or tightened to decrease their length. The pair of straps **136** can be located in a first, stowed position in which the straps **136** are tightened and the third deck panel **2794** and the fourth deck panel **2798** are substantially flush or flush with the top of the deck **2720**, such as the first deck panel **2792** and the second deck panel **2796** and in a second, deployed position in which the straps **136** are loosened and the third deck panel **2794** and the fourth deck panel **2798** are located a distance below the deck **2720**. FIGS. **4A** and **4B** show the straps **136** in the second, deployed position with the third deck panel **2794** and the fourth deck panel **2798** supported by the straps **136**.

[0113] According to at least some embodiments, the removable deck panels, such as for example the third deck panel **2794** and the fourth deck panel **2798**, are part of the collapsible footwell feature **102**. In particular, the wagon chassis **104** may include the collapsible footwell feature **102** and the removable deck panels, such as for example the third deck panel **2794** and the fourth deck panel **2798**, may be configured to transition from a first position, as shown in FIG. **1A**, and a second position, as shown in FIGS. **3A**, **3C**, **4A**, and **4B**. In the first position, the third deck panel **2794** and the fourth deck panel **2798** are flush with the first deck panel **2792** and the second deck panel **2796**. In the second position the third deck panel **2794** and the fourth deck panel **2798** are spaced a distance away from the first deck panel **2792** and the second deck panel **2796** to form the footwell **102**. For example, the collapsible footwell feature **102** is extended below the deck **2720**. The second position of the collapsible footwell feature **102** may be interchangeably referred to as a use position as a child sitting in a modular wagon bed **106** or on the chassis deck **2720** may extend their legs and feet into the collapsible footwell feature **102** in the user position for more comfort. In some embodiments, in the second position, the third deck panel **2794** and the fourth deck panel **2798** may be moved or slid as shown in FIG. **2C**, or removed entirely from the wagon chassis **104**.

Removable Pads

[0114] FIGS. **5A-5E** illustrate embodiments of an optional accessory, the optional accessory being one or more removable pads **140** that may be used with the wagon chassis **104** and modular wagon bed **106**. In these embodiments, the modular wagon bed **106** may be open along its bottom edge such that the deck **2720** forms the bottom of the wagon **100** when the modular wagon bed **106** is assembled to the wagon chassis **104**. The removable pads **140** includes one or more pegs **122** that cooperate with the openings **2808** in the deck **2720**, as described herein, to couple the removable pad **140** to the wagon chassis **104**. As shown in FIGS. **5A** and **5B**, in some embodiments, the removable pad **140** may be segmented and may be able to fold into a smaller size and/or may be folded while still coupled to the wagon chassis **104** as shown in FIG. **5B**. In some embodiments, the removable pad **140** as shown in FIG. **5A** is coextensive with the deck **2720**. In some embodiments, the removable pad **140** may be smaller than and cover less than the entire surface area of the top of the deck **2720**. For example, FIGS. **5C-5E** illustrate an embodiment that includes two smaller removable pads **140**, each having one or more pegs **122**. One of the removable pads **140** is configured to couple to the first deck panel **2792** and the other of the removable pads **140** is configured to couple to the second deck panel **2796**. In this embodiment, there is a space between the two removable pads **140** to permit the feet of one or more persons

riding in the wagon **100** to rest on the collapsible footwell **102** formed by the lowered third deck panel **2794** and fourth deck panel **2798**. The removeable pads **140** shown in FIGS. 5A-5E are cushioned and may provide a comfortable seating surface for persons who may ride in the wagon **100**. Also, as shown in FIGS. 5B and 5D, the removable pads **140** may remain coupled to the wagon chassis **104** when the wagon chassis **104** is folded.

Bag

[0115] FIGS. 6A-6C show the wagon chassis **104** with an optional accessory, wherein the optional accessory is a bag **302**. FIG. 6A shows an embodiment of the bag **302** on the wagon chassis **104** and FIG. 6B shows the bag **302** above the wagon chassis **104**. As shown in FIG. 6B, embodiments of the bag **302** include pegs **122** to aid in securing the bag **302** to the wagon chassis **104**. The pegs **122** on the bag are the same type of pegs **122** that are on the modular wagon bed **106**. In some embodiments, the bag **302** has a rigid bottom and/or the pegs **122** are rigidly connected to one another. In these embodiments, and where the wagon chassis **104** is foldable, the rigid bottom of the bag **302** or the rigid connection between the pegs **122** can keep the wagon chassis **104** in the unfolded position even if the wagon chassis **104** lacks a lock mechanism or device to keep the wagon chassis **104** in the unfolded position. As such, in these embodiments, the wagon chassis **104** can be easily folded without unlocking a lock mechanism or device by removing the bag **302** from the wagon chassis **104** and simply folding the wagon chassis **104**.

[0116] As shown in FIG. 6C, according to various embodiments, the wagon chassis **104** may include various removable or integrated securing elements **304** such as straps, bungees, buttons, clasps, ratchets, belts, or the like for securing an optional accessory, such as for example the bag **302**, to the wagon chassis **104** during transportation or storage. For example, the securing elements **304** may have a clip that is configured to clip onto the tie downs **2804** in the deck **2720** of the wagon chassis **104** (see FIGS. 1A, 1B). In another example, a rectangular ring may be mounted to the wagon chassis for supporting a cooler or other smaller containers. In other embodiments, an adapter (not shown) may be removably coupled to the wagon chassis **104** that may be removably coupled to an optional accessory with similar securing elements **304**. In other embodiments, any securing elements **304** or adapters may be snug fit to the optional accessory for securing the optional accessory to the wagon chassis **104** and vice versa.

Child Seat Adapters

[0117] FIGS. 7A-7C illustrate exemplary optional additional components that may be used with the wagon chassis **104**, including exemplary adapters **502** for use with the wagon chassis **104** to couple accessories **504** to the wagon chassis **104**. FIG. 7A illustrates a wagon chassis **104** having a seat adapter **502** for removably coupling a child seat **504** to the wagon chassis **104**. Various embodiments of a seat adapter **502** for a child seat **504** as discussed in U.S. Pat. No. 9,469,324 filed Jun. 25, 2015, U.S. Pat. No. 10,077,062 filed Oct. 3, 2016, U.S. Pat. No. 10,597,058 filed Sep. 17, 2018, U.S. Pat. No. 11,214,291 filed Feb. 19, 2020, and U.S. Pat. No. 11,787,460 filed Oct. 5, 2021, the entirety of which are hereby incorporated by reference.

[0118] One or more seat adapters **502** may be removably coupled to the wagon chassis **104** for coupling one or more corresponding child seats **504** as would be appreciated by one having ordinary skill in the art. As shown in FIGS. 7A and 7C, the seat adapters **502** include a bottom end having pegs **122** extending therefrom, that cooperate with the openings **2808** in the deck **2720**, as described herein, to couple the seat adapters **502** to the wagon chassis **104**. As shown in FIG. 7B, the seat adapters **502** do not include the pegs **122**, but include two latch components **506** that removably mount the seat adapters **502** to the frame **2716** of the wagon chassis **2716**. The latch components **506** shown in FIG. 7B may clamp to the frame **2716** of the wagon chassis **104**.

[0119] The adapters **502** further include seat latch components **508** that removably attach the seat adapters **502** to corresponding latch components **510** of the child seat **504**. The seat latch components **508** are manipulable between locked and unlocked positions, with the latches secured to the child seat **504** in the locked position (depicted) and released for detachment from the child

seat **504** in the unlocked position. One or more actuators **512** are operable to manipulate the latch components **510** between the locked and unlocked positions. In the depicted embodiment, the child-seat latch components **510** are of a conventional design determined by the child seat manufacturer, and the seat-adapter latch components **508** are of a conventional design selected for mating with the child-seat latch components (and thus typically conform to the design of the mating latch of the product the child seat is designed to be mounted to and used with). In other embodiments, the latch components are reversed with the actuator on the child seat and the adapter/seat latch component being passive. Details of such child seat latch components (e.g., catch-pins), and designs of mating latch components (e.g., pivotable hooks) and their actuators (e.g., spring-biased pushbuttons), are well-known in the art and thus not repeated for brevity's sake. [0120] As shown in FIG. 7B, the seat adapters **502** do not include the pegs **122**, but include two latch components **506** that removably mount the seat adapters **502** to the frame **2716** of the wagon chassis **2716**.

[0121] In various embodiments, the seat adapters **502** may be coupled to the wagon chassis **104** such that the child seats **504** face each other as shown in FIG. 7C. In other embodiments, the adapters **502** may be coupled to the wagon chassis **104** such that the child seats **504** face the same way (e.g., toward the handle **114** or away from the handle **114** as shown in FIGS. 7A and 7B) or facing opposing directions as described by the user. Any space created between the one or more child seats **504** may be used for additional storage according to various embodiments. Again, as shown in FIG. 7C for example, any of the foregoing embodiments may be used with the collapsible footwell feature **102** in either the first position or the second position as previously described.

Alternative Handles

[0122] FIGS. 8A-8E show a wagon chassis **104** with an alternative couplable pivotable handle **702**. According to other embodiments, the wagon chassis **104** may include a removably couplable pivotable handle **702**. As shown in FIGS. 8A-8C, the handle **702** may be a substantially U-shaped handle **702** with two legs **704** connected by a crossbar **706** extending between the two legs **704**. The two legs **704** are pivotably connected to the wagon chassis **104**. The handle **702** may pivot from a first position at one end of the wagon chassis **104** (e.g., an end proximate to the relatively larger wheels **112b** (as shown in FIG. 7A) for making it easier for a user to push the wagon chassis **104**) to second position at the opposing end of the wagon chassis (e.g., at an end proximate to the relatively smaller wheels **112a** (as shown in FIG. 7C) for making it easier for a user to pull the wagon chassis **104**). The handle **702** may be pivoted to and retained in one or more intermediate positions as shown in FIG. 7B. The handle **702** may be removably couplable to the wagon chassis **104** and may be position-locking according to any of the embodiments as described herein.

[0123] Now with reference to FIGS. 8D-8E, the handle **702** may be pivotably and removably coupled to the wagon chassis **104** for placement at either end of the wagon chassis **104**. For example, the handle **702** may be placed on an end proximate to relatively larger wheels **112b** (as shown in FIG. 8E) for making it easier for a user to push the wagon chassis **104** or the handle **702** may be placed on an end proximate to relatively smaller wheels **112a** (as shown in FIG. 8D) for making it easier for a user to pull the wagon chassis **104**. The handle **702** may be a position-locking handle feature. On a typical wagon the handle is free to rotate about its axis until its movement is limited by the wagon geometry or the ground. But on the wagon chassis **104** of the present disclosure, the handle **702** locks in one or more positions. Locking handle design options include automatic lock or only when intentioned by the user. Additionally, in some embodiments, the handle **702** is sized and shaped such that the handle **702** can be folded flat for storage. In such embodiments, the legs **704** are of sufficient length and the crossbar **706** is of sufficient width to permit the handle **702** to encircle three sides of the frame **2716** of the wagon chassis **104** when the handle **702** is folded flat. Stated another way, when the handle **702** is coupled to an end of the wagon chassis **104**, the handle **702** is sized to be around at least a portion of the perimeter of the frame **2716** of the wagon chassis **104** when the handle **702** is folded toward the frame **2716**.

Wagon Chassis with Center Support

[0124] FIG. 9 shows another embodiment of a wagon chassis **800**. According to this embodiment, the wagon chassis **800** includes a handle **114** coupled to a center support **804** supported by one a first pair of wheels **112a** and a second pair of wheels **112b**. The wheels **112a**, **112b** may each be coupled to frame cross members **808** extending from the center support **804**. In some embodiments, the frame cross members **808** may include a spring or other dampening element to reduce any bounce experienced by optional accessories coupled to the center support **804** during transport. The wheels **112a**, **112b** may include any configuration (e.g., diameter, width, size, etc.) as described above with respect to other figures for traversing a variety of terrains. The wagon chassis **800** further includes the deck **2720**, which may be coupled to the center support **804**. In some embodiments, the deck **2720** is removably coupled to the center support **804**. As described herein, the deck **2720** includes one or more of the first deck panel **2792**, second deck panel **2796**, third deck panel **2794**, and fourth deck panel **2798**. The modular wagon bed **106** may be coupled to the wagon chassis **800** by inserting the pegs **122** of the modular wagon bed **106** into the openings **2808** of the deck **2720**.

Retractable Cargo Deck

[0125] FIGS. 10A-10C illustrate an embodiment of the wagon chassis **104** that includes a retractable cargo deck **1100**. The retractable cargo deck **1100** may transition between a first, storage position as shown in FIG. 10A and a second, extended position as shown in FIGS. 10B and 10C. In various embodiments, the retractable cargo deck **1100** may be slidable or foldable under the wagon chassis **104**. For example, the retractable cargo deck **1100** may be connected to the wagon chassis **104** via slidable or telescoping rails **1102**. In some embodiments, the retractable cargo deck **1100** also includes attachment features **2802**, such as tie downs **2804** and openings **2808**, as on deck **2720**. In some embodiments, the retractable cargo deck **1110** does not include any attachment features **2802**, such as tie downs **2804** and openings **2808**. As shown in FIG. 10C, an optional accessory, such as for example a cooler **1104**, may be placed on an carried by the retractable cargo deck **1100**.

Adapters

[0126] FIGS. 11A-11B illustrate an exemplary adapter. Exemplary adapter **1200** may be a rack for sliding an accessory or a portion of an accessory, such as folded beach chairs or the like, over. For example, an adapter **1200** incorporating a rack may be used to carry chairs, boogie boards, sports equipment, skateboards, or the like. The adapter **1200** shown in FIGS. 11A-11B, for example, is a rack **1202** that is U-shaped having a proximal end and a distal end, with the distal end angled at 90 degrees with respect to the proximal end. Namely, the rack **1202** includes a two legs **1204** connected by a crossbar **1206** extending between the two legs **1204**. The two legs **1204** have a 90 degree bend in them to provide a portion of the legs **1204** that are substantially parallel or parallel to the frame **2716** of the wagon chassis **104**. Additionally, as shown in FIGS. 11A-11B, each of the two legs **1204** includes a wagon latch component **510**, as described with respect to FIG. 7B. Although FIGS. 11A-11B illustrate the adapter **1200** being coupled to the modular wagon bed **106**, the adapter **1200** may be similarly removably coupled to the wagon chassis **104**. Any adapter described herein may be removably coupled to any wall or end of the modular wagon bed **106** or the wagon chassis **104**.

[0127] FIGS. 12A-12B illustrate another exemplary optional adapter **1600** that may be coupled to the wagon chassis **104**. For example, the adapter **1600** shown in FIGS. 12A-12B includes a frame **1602** and one or more straps **1606**. The frame **1602** is substantially U-shaped with two upright legs **1608** connected by a top bar **1610** extending between the two legs **1608**. Each of the two upright legs **1608** includes a peg **122**, which cooperate with the openings **2808** of the deck **2720** of the wagon chassis **104** to connect the adapter **1600** to the wagon chassis **104**. Thus, the adapter **1600** may be coupled to the wagon chassis **104** by inserting the pegs **122** of the adapter **1600** into the openings **2808** of the deck **2720**. The top bar **1610** may be at an angle with respect to the upright

legs **1608**. The top bar **1610** may be used as a handle in some configurations. As shown in FIG. **12B**, the adapter **1600** is shown coupled to the wagon chassis **104** along with two optional accessories coupled to the adapter **1600**. The optional accessories may be for example, foldable sports chairs, foldable tables, a tent in a bag, etc. The optional accessories are shown resting on the optional retractable cargo deck **1100** and may be secured to the frame **1602** by the one or more straps **1606**.

[0128] FIGS. **13A** and **13B** illustrate another exemplary adapter **1700** that may be coupled to the wagon chassis **104**. For example, adapter **1700** is a fishing pole adapter. Adapter **1700** includes one or more tube members **1702** coupled to a frame **1704**. The frame **1704** is substantially U-shaped with two upright legs **1708** connected by a first cross bar **1710** extending between the two legs **1708**. The frame **1704** may also include a second cross bar **1712**. Each of the two upright legs **1708** includes a peg **122**, which cooperate with the openings **2808** of the deck **2720** of the wagon chassis **104** to connect the adapter **1700** to the wagon chassis **104**. Thus, the adapter **1700** may be coupled to the wagon chassis **104** by inserting the pegs **122** of the adapter **1700** into the openings **2808** of the deck **2720**. The tube members **1702** are coupled to first cross bar **1710** and the second cross bar **1712**, where provided. The one or more tube members **1702** are configured to receive a fishing pole, a spear, a golf club, or any other long and narrow item. In other embodiments, the diameter of the tube member **1702** may be greater to accommodate thicker items such as baseball bats, tennis rackets, liquid containers, etc. In some embodiments, the tube members **1702** include slots **1714** that are sized and/or shaped to receive within the slots **1714**, for example, the reel of a fishing rod. In at least some embodiments, the adapter **1700** may include a variety of sizes (e.g., diameter and/or length) of tube members **1702**. In further embodiments, the tube members **1702** may be interchangeable such that a user may select the size of the tube member **1702** according to their intended use for the day (e.g., baseball practice versus a fishing day).

[0129] FIGS. **14A**, **14B**, and **15A-15D** illustrate another exemplary adapter **1900** that may be coupled with the wagon chassis **104**. For example, adapter **1900** may be another rack configuration suitable for transporting accessories such as chairs (such as shown in FIGS. **14B** and **15A**), bikes (such as shown in FIG. **15C**), snowboards (such as shown in FIG. **15D**), boogie boards (such as shown in FIG. **15B**), etc. One or more accessories may be used with the adapter **1900** which may be the same type or a variety of accessories. The adapter **1900** includes a generally upright leg **1902**, a cross bar **1904** coupled to the top of the leg **1902**, and one or more arms **1906** extending from the cross bar **1904**. The terminal ends of the arms **1906** may be bent slightly to aid in retaining any accessories that are used with the adapter **1900**. The upright leg **1902** and the cross bar **1904** form a generally T-shaped frame. The adapter **1900** further includes pegs **122** (as shown in FIG. **14B**), which cooperate with the openings **2808** of the deck **2720** of the wagon chassis **104** to connect the adapter **1900** to the wagon chassis **104**. Thus, the adapter **1900** may be coupled to the wagon chassis **104** by inserting the pegs **122** of the adapter **1900** into the openings **2808** of the deck **2720**. When the adapter **1900** is coupled to the wagon chassis **104**, the arms **1906** may be generally parallel to the deck **2720** of the wagon chassis **104**. In some embodiments, the leg **1902** is a telescopic leg that can be extended or retracted, which may aid in carrying accessories of different lengths and/or aids in storing the adapter **1900** when not in use.

[0130] The arms **1906** may be removable from the adapter **1900** and the number of arms **1906** coupled to the cross bar **1904** can be increased or decreased depending on the accessories **1000** that a user desires to carry. Additionally, the spacing between each of the arms **1906** may be increased or decreased depending on the accessories **1000** that a user desires to carry. For example, FIG. **14B** illustrates the adapter **1900** with two arms **1906** and FIGS. **14A** and **15A-15D** illustrate the adapter **1900** with four arms **1906**.

Removable and Relocatable Wheels

[0131] Referring again to FIG. **1A** and FIGS. **16**, **17A**, and **17B**, in some embodiments, the location of the wheels **112a** and **112b** on the wagon chassis **104** can be changed. In FIG. **1A**, the swivel

wheels **112a** are shown coupled to the first swivel wheel housing **2726** and the second swivel wheel housing **2730** proximate to the handle **114** and the wheels **112b** are shown coupled to the frame **2716** proximate to the second frame member **2728**. In this configuration, the swivel wheels **112a** may be considered at the front of the wagon chassis **104** and the wheels **112b** may be considered at the rear of the wagon chassis **104**. Additionally, in this configuration, the wagon chassis **104** may be pulled like a traditional wagon.

[0132] As shown in FIG. **16**, the swivel wheels **112a** include stems **1910**. The stems **1910** of the swivel wheels **112a** may be removed from the sockets of the first swivel wheel housing **2726** and the second swivel wheel housing **2730** and the wheels **112b** may be removed from the frame **2716**. In some embodiments, the wheels **112b** are a part of a wheel assembly **1912** that includes a pair of wheel mounts **1914** and an axle **1916** coupled to and extending between the wheel mounts **1914**. The wheels **112b** are rotatably coupled to the axle **1916**, which is fixedly coupled to the wheel mounts **1914**. In some embodiments, the wheels **112b** are fixedly coupled to the axle **1916** and the axle **1916** is rotatably coupled to the wheel mounts **1914**. The wheel assembly **1912** is coupled to the frame **2716** of the wagon chassis **104**, for example, by two threaded fasteners **1918** extending through each of the wheel mounts **1914** and into corresponding threaded holes in the frame **2716**.

[0133] As shown in FIGS. **17A** and **17B**, the location of the wheels **112a** and the wheels **112b** is swapped with respect to the location shown in FIGS. **1A** and **16**. In FIG. **17A**, the wheels **112b** are shown coupled to the frame proximate to the handle **114** and the swivel wheels **112a** are shown coupled to the third swivel wheel housing **2734** and the fourth swivel wheel housing **2738** proximate to the second frame member **2728**. For example, the stems **1910** of the swivel wheels **112a** are in the sockets of the third swivel wheel housing **2734** and the fourth swivel wheel housing **2738** and the wheel assembly **1912** is coupled to the frame **2716** of the wagon chassis **104**, for example, by two threaded fasteners **1918** extending through each of the wheel mounts **1914** and into corresponding threaded holes in the frame **2716** proximate to the first frame member **2724**. In this configuration, the wagon chassis **104** may be pushed, for example, like a stroller.

Additional Embodiment of Wagon Chassis

[0134] Referring now to FIGS. **18A-18D**, another exemplary embodiment of the wagon chassis **104** is shown. As shown in FIGS. **18B-18C**, in some embodiments, the frame **2716** of the wagon chassis **104** further includes a cross member **2784** that is coupled to and extends from the first hinge **2756** to the second hinge **2760**. The cross member **2784** provides structural strength and/or rigidity to the wagon chassis **104**. In some embodiments, the wagon chassis **104** additionally or alternatively includes a cross member extending between the third frame member **2732** and the fifth frame member **2740** and/or a cross member extending between the fourth frame member **2736** and the sixth frame member **2744**.

[0135] With reference continued to FIGS. **18B-18C**, in some embodiments, the wagon chassis **104** further includes a fold actuator **2788** that is configured to be actuated to unlock the first hinge **2756** and/or the second hinge **2760** to permit folding of the wagon chassis **104**. For example, the fold actuator **2788** is a slide actuator or a rotatable actuator that pulls on cables (not shown) that pull pins in each of the first hinge **2756** and the second hinge **2760** that hold the first hinge **2756** and the second hinge **2760** in the first, unfolded, or use position. As shown, the fold actuator **2788** is located on the cross member **2784**. However, in other embodiments, one or more fold actuators **2788** may be located on other portions of the wagon chassis **104** such as, for example without limitation, on one or more of the first, second, third, fourth, fifth, and/or sixth frame members **2724**, **2728**, **2732**, **2736**, **2740**, **2744**.

[0136] Now with reference to FIGS. **18A** and **18B**, in some embodiments, the deck **2720** of the wagon chassis **104** further includes an optional fold plate **2812** connected to the second deck panel **2796**. The fold plate **2812** may be connected to the second deck panel **2796** by one or more hinges (not shown) and may be spring loaded to keep the fold plate **2812** flat against the first deck panel **2792** and the second deck panel **2796**. It will be understood that in some embodiments, the fold

plate **2812** may be connected to the first deck panel **2792** without departing from the scope of the disclosure.

[0137] When a user desires to fold up the wagon chassis **104**, the user can flip up the fold plate **2812**, revealing an opening between the first deck panel **2792** and the second deck panel **2796**. The user may then be able access the fold actuator **2788** to fold the wagon chassis **104**.

Storage Devices

[0138] Now with reference to FIGS. **19A-19C**, the foldable modular wagon **100** includes the wagon chassis **104**. As shown in FIGS. **19A-19C**, in some embodiments, the modular wagon bed **106** includes an optional accessory, the optional accessory being one or more storage devices, such as for example only, two bags **2816** coupled to the wagon chassis **104**. As shown in FIG. **19A**, embodiments of the bags **2816** include pegs **122** to aid in securing the bags **2816** to the wagon chassis **104**. The pegs **122** on the bags **2816** are the same type of pegs **122** that are on the modular wagon bed **106**. For example, a first of the two bags **2816** is coupled to the first deck panel **2792** and the third deck panel **2794** and a second of the two bags **2816** is coupled to the second deck panel **2796** and the fourth deck panel **2798**. The two bags **2816** may themselves be collapsible or foldable bags and are shown in their first, uncollapsed, or use position in FIG. **19A** and in their second, collapsed, or storage position in FIGS. **19B** and **19C**. As shown in FIG. **19C**, the wagon chassis **104** is shown in its second, folded, or storage position and the bags **2816** can remain coupled to the wagon chassis **104** when it is in second, folded, or storage position. Additionally, the handle **114** can be folded down toward the first deck panel **2792** to sandwich the storage device (e.g., one of the bags **2816**) against the first deck panel **2792** and the third deck panel **2794**.

Additional Embodiments of Deck

[0139] Now with reference to FIGS. **20A-20C**, embodiments of the deck **2720** are shown. As shown in FIG. **20A**, the deck **2720** comprises a single deck panel **2722**. As shown in FIG. **20B**, the deck **2716** comprises two deck panels **2722**. As shown in FIG. **20C**, the deck **2716** comprises four deck panels **2722**. Additionally, as shown in FIGS. **20A-20C**, the deck panels **2722** may be removable from the wagon chassis **104**. Additionally, as shown in FIGS. **20A-20C**, in some embodiments, the wagon chassis **104** is not foldable.

[0140] Another embodiment of the wagon chassis **104** is shown in FIG. **21**. In some embodiments, the wagon chassis **104** includes fold out deck panels that increase the surface area of the deck **2720** of the wagon chassis **104**. The increased surface area of the deck **2720** may permit carrying more and/or larger items on the wagon chassis **104**. For example, the wagon chassis **104** includes a first folding deck panel **2100** and a second folding deck panel **2102**, wherein each of the first folding deck panel **2100** and the second folding deck panel **2102** is rotatably coupled to the wagon chassis **104** by one or more hinges **2104**. The hinges **2104** are located on the left side and right side of the wagon chassis **104**. For example, one or more hinges **2104** are coupled to the third frame member **2732** (not visible in FIG. **21**) and the fourth frame member **2736** (not visible in FIG. **21**) on the left side of the wagon chassis **104** and one or more hinges **2104** are coupled to the fifth frame member **2740** and the sixth frame member **2744** on the right side of the wagon chassis **104**. Each of the first folding deck panel **2100** and the second folding deck panel **2102** may be selectively, and independently of one another, positioned in a first, folded position and a second, unfolded position. Each of the first folding deck panel **2100** and the second folding deck panel **2102** have a first surface **2106** and a second surface **2108**, opposite the first surface **2106**.

[0141] In the first, folded position, the first folding deck panel **2100** and the second folding deck panel **2102** are located on top of the deck **2720** such that the first surface **2106** of each of the first folding deck panel **2100** and the second folding deck panel **2102** face the top surface **2782** of the deck **2720** and are proximate to and substantially parallel or parallel to the top surface **2782** of the deck **2720**. Additionally, in the first, folded position, the second surface **2108** of each of the first folding deck panel **2100** and the second folding deck panel **2102** face upward and form a top surface upon which items may be placed. In the second, unfolded position, the first folding deck

panel **2100** and the second folding deck panel **2102** are folded 180 degrees from the first, folded position outward on the left side and the right side, respectively, of the wagon chassis **104**, and are cantilevered and extend outward. Additionally, in the second, unfolded position, the first surface **2106** of each of the first folding deck panel **2100** and the second folding deck panel **2102** face upward and form a top surface upon which items may be placed, and the second surface **2108** of each of the first folding deck panel **2100** and the second folding deck panel **2102** face downward. [0142] As shown in FIG. **21**, one or both of the first folding deck panel **2100** and the second folding deck panel **2102** include attachment features **2802**, such as tie downs **2804** and openings **2808**, as on deck **2720**. In some embodiments, one or of the first folding deck panel **2100** and the second folding deck panel **2102** do not include any attachment features **2802**, such as tie downs **2804** and openings **2808**. In some embodiments, one or both of the first folding deck panel **2100** and the second folding deck panel **2102** are removable from the wagon chassis **104**.

Telescoping Wagon Chassis

[0143] In some embodiments, as shown in FIGS. **22A-22C**, the wagon chassis **104** may be adjustable in length to shorten or lengthen the wagon chassis **104**. For example, the wagon chassis **104** may be lengthened to carry longer items on the wagon chassis **104** or it may be shortened to carry shorter items on the wagon chassis **104**, or for storage of the wagon chassis **104**.

[0144] As shown in FIGS. **22A-22C**, the frame **2716** of the wagon chassis **104** is telescopic. For example, the third frame member **2732** and the fifth frame member **2740** are configured to be received in the fourth frame member **2736** and the sixth frame member **2744**, respectively, and the third frame member **2732** and the fifth frame member **2740** are configured to slide at least partially into and out of the fourth frame member **2736** and the sixth frame member **2744**, respectively. The third frame member **2732** and the fifth frame member **2740** may be sized and/or shaped to fit within in the fourth frame member **2736** and the sixth frame member **2744**, respectively.

Additionally, as shown in FIGS. **22A-22C**, the deck **2720** is telescopic. For example, the deck **2720** may include the first deck panel **2792** and the second deck panel **2796**, wherein the second deck panel **2796** is mounted higher on the frame than the first deck panel **2792** so that the second deck panel **2796** can overlap at least a portion of the first deck panel **2792** and slide with respect to the first deck panel **2792**.

[0145] Thus to shorten the wagon chassis **104**, for example, the first frame member **2724** and the second frame member **2728** of the may be brought to toward one another, with the third frame member **2732** and the fifth frame member **2740** telescoping within at least a portion of the fourth frame member **2736** and the sixth frame member **2744**, respectively, and the second deck panel **2796** sliding over at least a portion of the first deck panel **2792**. To lengthen the wagon chassis **104**, this process is reversed. In some embodiments, the wagon chassis **104** may be locked its short length (see, e.g., FIG. **22A**), in its standard length (see, e.g., FIG. **22B**), its long length (see, e.g., FIG. **22C**), or any intermediate length between these lengths using lock devices, such as for example, quick-release clamps, ratchet systems, screws, a series of spring-loaded pins, etc.

[0146] In some embodiments, for example, the above exemplary arrangement is reversed, wherein the fourth frame member **2736** and the sixth frame member **2744** are configured to be received in the third frame member **2732** and the fifth frame member **2740**, respectively, and the fourth frame member **2736** and the sixth frame member **2744** are configured to slide at least partially into and out of the third frame member **2732** and the fifth frame member **2740**, respectively. In this configuration, the first deck panel **2792** is mounted higher on the frame than the second deck panel **2796** so that the first deck panel **2792** can overlap at least a portion of the second deck panel **2796** and slide with respect to the second deck panel **2796**.

Corner Block Adapters

[0147] FIGS. **23A-23E** illustrate another exemplary adapter. In FIGS. **23A-23E**, the adapter comprises one or more corner blocks **2300** that may be coupled to the deck **2720** of the wagon chassis **104** to carry items of different and/or irregular sizes. The corner blocks **2300** can be

mounted anywhere on the deck **2720** as needed or desired to carry an item.

[0148] Referring now to FIGS. **23B** and **23C**, an exemplary corner block **2300** is shown. The corner block **2300** includes a body **2302**, one or more resilient members **2304**, one or more pegs **122**, a support surface **2306**, and one or more tie downs **2308**.

[0149] The body **2302** has a top side **2310**, a bottom side **2312**, and a height **2314** between the top side **2310** and the bottom side **2312**. Coupled to and extending from the bottom side **2312**, and away from the top side **2310** are the one or more resilient members **2304** and the one or more pegs **122**, with a first end of the resilient member **2304** coupled to the bottom side **2312** and a second end of the resilient member **2304** connected to the peg **122**. As such, the resilient member **2304** is located between the body **2302** and the peg **122**.

[0150] The one or more resilient members **2304** may include, but are not limited to, an elastomeric rubber, a coil spring, a fluid-filled damper, etc. For example, the elastomeric rubber may be overmolded onto and between the peg **122** and the body **2302** to elastically couple the body **2302** and the peg **122**. As another example, a coil spring may be coupled to the body **2302** and the peg **122** and a hollow bellows cover may be provided around the coil spring.

[0151] The support surface **2306** is located between the top side **2310** and the bottom side **2312** of the body **2302** and at a depth **2316** from the top side **2310**. The support surface **2306** is square shaped, although other shapes may be provided without departing from the scope of the disclosure. The body **2302** further includes two sidewalls **2318**, with each sidewall **2318** extending upward from the support surface **2306** toward the top side **2310** and oriented perpendicular to the support surface **2306**, the top side **2310**, and the other sidewall **2318**. The support surface **2306** and the two sidewalls **2318** cooperate to form a cube-shaped recess in the body **2302**.

[0152] The tie downs **2308** are located on the periphery of the body **2302** opposite the support surface **2306**, with a sidewalls **2318** of the respective sidewalls **2318** between the support surface **2306** and a tie down **2308** of the respective tie downs **2308**. Each tie down **2308** includes a bar **2320** located in a recess **2322** in the body **2302**. However, it should be understood that in some embodiments, for example only, the tie downs **2308** may lack the recess and the bars may be U-shaped bar extending upward from the top side **2310** of the body **2302**. As shown in FIGS. **23B** and **23C**, the bars **2320** are parallel to the top side **2310** of the body **2302**, and each bar **2320** is perpendicular to one the other bar **2320**.

[0153] Referring again to FIG. **23A** and to FIGS. **23D** and **23E**, the corner blocks **2300** are shown coupled to the wagon chassis **104** along with a cargo bin **2324** supported by the corner blocks **2300**. The pegs **122** of each corner block **2300** are inserted into respective openings **2808** in the smart grid on the deck **2720** of the wagon chassis **104**. The resilient member **2304** is thus located between the deck **2720** and the support surface **2306** to provide a level of vibration and movement isolation between the wagon chassis **104** and the cargo bin **2324**. As such, the resilient member **2304** acts like a suspension system to reduce or eliminate damage to fragile items that may be carried on the wagon chassis **104**, such as items within the cargo bin **2324**. Additionally, the corners of the cargo bin **2324** are received in the cube-shaped recess formed by the support surface **2306** and the sidewalls **2318** in the body **2302**. The bottom of the cargo bin **2324** is supported by the support surface **2306** of the corner block **2300**. The support surfaces **2306** and the sidewalls **2318** of the corner blocks **2300** can thus retain an item, such as the cargo bin **2324**, on the wagon chassis **104**. Additionally, as shown in FIGS. **23A**, **23D**, and **23E**, one or more securing elements **304** such as bungee cords, ropes, ratchet straps, or the like may be clipped or tied to the tie downs **2308** to secure the item, such as the cargo bin **2324**, on the wagon chassis **104**.

[0154] Referring now to FIGS. **24A** and **24B**, an alternative embodiment of the corner block **2300** is shown that does not include the resilient members **2304**. In the embodiment of corner block **2300** shown in FIGS. **24A** and **24B**, the one or more pegs **122** are coupled directly to the bottom side **2312** of the body **2302** of the corner block **2300**.

Tie Down Blocks

[0155] FIGS. 25A-25D illustrate another exemplary adapter. In FIGS. 25A-25D, the adapter comprises one or more tie down blocks **2500** that may be coupled to the deck **2720** of the wagon chassis **104** to carry items of different and/or irregular sizes. The tie down blocks **2500** can be mounted anywhere on the deck **2720** as needed or desired to carry an item.

[0156] Referring now to FIG. 25B, an exemplary tie down block **2500** is shown. The tie down block **2500** includes a body **2502**, one or more pegs **122**, and one or more tie downs **2504**. The body **2502** has a top side **2506** and a bottom side **2508**. Coupled to and extending from the bottom side **2508**, and away from the top side **2506** is a peg **122**. The tie down **2504** includes a bar **2510** located in a recess **2512** in the body **2502**. However, it should be understood that in some embodiments, for example only, the tie downs **2504** may lack the recess and the bars may be U-shaped bar extending upward from the top side **2506** of the body **2502**.

[0157] Referring again to FIGS. 25A and 25D, the tie down blocks **2500** are shown coupled to the wagon chassis **104** along with, for example, a cooler **1104**. The pegs **122** of each tie down block **2500** are inserted into respective openings **2808** in the smart grid on the deck **2720** of the wagon chassis **104**. Additionally, as shown in FIGS. 25A and 25D, one or more securing elements **304** such as bungee cords, ropes, ratchet straps, or the like may be clipped or tied to the tie downs **2504** to secure the item, such as the cooler **1104**, on the wagon chassis **104**.

Adjustable Mounting Rails

[0158] FIGS. 26A and 26B illustrate another exemplary adapter. In FIGS. 26A and 26B, the adapter comprises one or more mounting rails **2600** that may be coupled to the deck **2720** of the wagon chassis **104** to carry items of different and/or irregular sizes. The mounting rails **2600** can be mounted anywhere on the deck **2720** as needed or desired to carry an item.

[0159] The mounting rails **2600** include a rail **2602**, two or more adjustable blocks **2606**, and two or more pegs **122**. The rail **2602** includes a channel **2604** on its top side in which the adjustable blocks **2606** are retained and are able to slide along the length of the rail **2602**. The two or more pegs **122** are coupled to and extend from the bottom side of the rail **2602**. The two or more pegs **122** are proximate to the terminal ends of the rail **2602**. Each adjustable block **2606** includes a clamping surface **2608**, which in some embodiments, is a rubber material. The clamping surface **2608** is substantially flat and is oriented perpendicular to the length of the rail **2602** when the adjustable block **2606** is mounted to the rail **2602**. Each adjustable block **2606** further includes a releasable lock **2610** to permit locking the adjustable block **2606** in a location along the length of the rail **2602**. In some embodiments, for example, the releasable lock **2610** is similar to a quick release lock, and includes a lever **2612** that is raised to release the releasable lock **2610** and lowered to lock the releasable lock **2610**. As shown in FIGS. 26A and 26B, the adjustable blocks **2606** are mounted to the rail **2602** such that the clamping surfaces **2608** of the adjustable blocks **2606** face one another. This permits clamping an item between the clamping surfaces **2608** of the adjustable blocks **2606**.

[0160] To use the mounting rails **2600**, a user can couple the mounting rails **2600** to desired locations on the deck **2720** of the wagon chassis **104** by inserting the pegs **122** of each mounting rail **2600** into respective openings **2808** in the smart grid on the deck **2720** of the wagon chassis **104**. As shown in FIG. 26A, the user can orient the mounting rails **2600** so that they are oriented transversely on the deck **2720** (e.g., wherein the rails **2602** extend left and right on the deck **2720** and are perpendicular to the left side and the right side of the wagon chassis **104**). Additionally, the user can space the mounting rails **2600** longitudinally (e.g., fore and aft) on the deck **2720** such that the item that the user desires to carry on the wagon chassis **104** is supported by the rails **2602** of the mounting rails **2600**. For example, as shown in FIG. 26A, where the item is a cooler **1104**, the user can space the mounting rails **2600** apart a distance that is less than the length of the cooler **1104** so that the bottom of the cooler **1104** rests on the rails **2602** of the mounting rails **2602**. The user can then unlock or release the releasable locks **2610** of the adjustable blocks **2606**, move the adjustable blocks **2606** so that the clamping surfaces **2608** contact the item that the user desires to securely

hold on the wagon chassis **104**, and can then lock the releasable locks **2610** of the adjustable blocks **2606** to secure the adjustable blocks **2606** in place. The user can compress the clamping surfaces **2608** of the adjustable blocks **2606** against the item, such as for example, the cooler **1104** or other items, such as for example, a boogie board, to secure the items to the wagon chassis **104**.

[0161] Although the mounting rails **2600** are shown oriented transversely on the deck **2720** in FIG. **26A**, it should be understood that the mounting rails **2600** may be oriented longitudinally on the deck **2720** as shown in FIG. **26C** (e.g., wherein the rails **2602** extend fore and aft on the deck **2720** and are parallel to the left side and the right side of the wagon chassis **104**).

Pegs on the Additional Components

[0162] As shown throughout the figures, in some embodiments, the pegs **122** described herein have a hexagonal shape. Now with reference to FIGS. **27A-27D**, various embodiments of pegs **122** are shown having alternative shapes and/or features. In FIG. **27A**, for example, the peg **122** includes threads **2614** and is configured to be screwed into a corresponding threaded opening **2808** in the deck **2720** of the wagon chassis **104**. In FIG. **27B**, for example, the peg **122** includes one or more ball locks **2616** that are configured to be received in one or more corresponding detents in openings **2808** in the deck **2720** of the wagon chassis **104**. Additionally, the tie down block **2500** includes a sprung push button **2618** that may be depressed to release the ball locks **2616** and permit removal of the peg **122** from the openings **2808** in the deck **2720**. In FIG. **27C**, for example, the peg **122** includes a pair of flex fingers **2620**, each having a retention flange **2622** at the distal end of the peg **122**. The pair of flex fingers **2620** are configured to flex toward one another to aid in inserting the peg **122** into the openings **2808** in the deck **2720**. In FIG. **27D**, the peg **122** includes a spring button clip fastener **2626** that is configured to be received in one or more corresponding detents in openings **2808** in the deck **2720** of the wagon chassis **104**. In some embodiments, the pegs **122** may also include one or more magnets and may be retained magnetically in the openings **2808** in the deck **2720**. For example, the deck **2720** may include or be made of a magnetic material (e.g., ferrous metal, a plastic magnet) to which the one or more magnets in the pegs **122** are attracted to aid in retaining the pegs **122** in the openings **2808** in the deck **2720**. In some embodiments, at least the internal surface(s) of the openings **2808** in the deck **2720** may be rubberized and the pegs **122** may be retained in the openings **2808** by a friction fit. Although FIGS. **27A-27D** show these pegs **122** in conjunction with a tie down block **2500**, it should be understood that the pegs **122** shown in FIGS. **27A-27D** can be used with any embodiment of the additional components (e.g., the accessories and adapters) described herein as including a peg **122** without departing from the scope of the disclosure.

Open Top Bin

[0163] FIG. **28** shows the wagon chassis **104** with an optional accessory, wherein the optional accessory is an open top bin **2800**. As shown in FIG. **28**, embodiments of the open top bin **2800** include pegs **122** to aid in securing the open top bin **2800** to the wagon chassis **104**. The pegs **122** on the open top bin **2800** are the same type of pegs **122** that are on the modular wagon bed **106**. Like the bag **302** described with respect to FIGS. **6A-6C**, in some embodiments, the open top bin **2800** has a rigid bottom and/or the pegs **122** are rigidly connected to one another. In these embodiments, and where the wagon chassis **104** is foldable, the rigid bottom of the open top bin **2800** or the rigid connection between the pegs **122** can keep the wagon chassis **104** in the unfolded position even if the wagon chassis **104** lacks a lock mechanism or device to keep the wagon chassis **104** in the unfolded position. As such, in these embodiments, the wagon chassis **104** can be easily folded without unlocking a lock mechanism or device by removing the open top bin **2800** from the wagon chassis **104** and simply folding the wagon chassis **104**.

Trailer Configurations

[0164] Now with reference to FIGS. **29A, 29B, 30A, 30B, 31, 32A, and 32B**, in some embodiments, the wagon chassis **104** may be configured as a trailer **2900**. When configured as a trailer, the swivel wheels **112a** are removed from the wagon chassis **104** and can be stored on the

deck **2720**, for example, by inserting the stems **1910** of the swivel wheels **112a** into the openings **2808** in the deck **2720**. Additionally, a tow bar **2904** is coupled to the wagon chassis **104** and the tow bar **2904** may be coupled to a vehicle, such as for example only, a bicycle **2902**.

[0165] As shown in FIGS. **29A** and **29B**, the tow bar **2904** includes a first end **2906** configured to be coupled to the wagon chassis **104** and a second end **2908** configured to be coupled to the bicycle **2902**. For example, the first end **2906** of the tow bar **2904** is coupled to the frame **2716** of the wagon chassis **104** by two threaded fasteners **1918** and the second end **2908** of the tow bar **2904** couples to, for example, the chain stay of the bicycle **2902** as is known in the art.

[0166] In FIGS. **30A** and **30B**, another embodiment of the wagon chassis **104** configured as a trailer **2900** is shown. The tow bar **2904** cooperates with the second handle portion **114b** to form a goose-neck style tow bar. As shown in FIG. **30A**, the first handle portion **114a** can be removed from the wagon chassis **104**. As shown in FIG. **30B**, the first end **2906** of the tow bar **2904** can be inserted into the second handle portion **114b**. The quick release clamp **114d** on the second handle portion **114b** is then clamped to secure the tow bar **2904** to the second handle portion **114b**. The second end **2908** of the tow bar **2904** is then clamped onto the seat tube **2910** of the bicycle **2902**.

[0167] In FIG. **31**, another embodiment of the wagon chassis **104** configured as a trailer **2900** is shown. In this embodiment, the tow bar **2904** couples to the first handle portion **114a** of the wagon chassis **104** and the seat tube **2910** of the bicycle **2902**. Namely, the first end **2906** of the tow bar **2904** is clamped to the first handle portion **114a** of the wagon chassis **104** and the second end **2908** of the tow bar **2904** is clamped to the seat tube **2910** of the bicycle **2902**.

[0168] In FIGS. **32A** and **32B**, another embodiment of the wagon chassis **104** configured as a trailer **2900** is shown. In this embodiment, the tow bar **2904** includes a cross bar **2912** at the first end **2906** of the tow bar **2904**. The cross bar **2912** includes a pair of stems **2914** extending upwardly from the cross bar **2912**. The stems **2914** are spaced apart a distance equal to the distance between the first swivel wheel housing **2726** and the second swivel wheel housing **2730** (and the distance between the third swivel wheel housing **2734** and the fourth swivel wheel housing **2738**). As shown in FIG. **32B**, the first end **2906** of the tow bar **2904** is coupled to the frame **2716** of the wagon chassis **104** by inserting the two stems **2914** into the sockets of the first swivel wheel housing **2726** and the second swivel wheel housing **2730** and the second end **2908** of the tow bar **2904** couples to, for example, the chain stay of the bicycle **2902** as is known in the art. As shown in FIGS. **32A** and **32B**, in some embodiments, the cross bar **2912** includes a downwardly extending portion to provide clearance between the cross bar **2912** and a portion of the handle **114** at the location where the handle **114** is coupled to the frame **2716** of the wagon chassis **104**.

[0169] Although in FIGS. **29A**, **29B**, **30A**, **30B**, **31**, **32A**, and **32B** the wagon chassis **104** is shown as being used in a trailer configuration and coupled to a bicycle **2902**, it should be understood that the wagon chassis **104** in the trailer configuration may be coupled to any type of vehicle, such as for example and without limitation, a motorcycle, an all-terrain vehicle (ATV), a side-by-side, a snowmobile, a car, a truck, a child's ride on toy vehicle (e.g., a pedal car, a battery-powered ride on vehicle (such as those sold under the Power Wheels™ brand), a go kart, etc.), a tricycle, etc. without departing from the scope of the disclosure.

Embodiments of Hinges

[0170] Referring now to FIGS. **33A-33D**, **34A**, and **34B**, an embodiment of locking hinges **2756**, **2760** of the wagon chassis **104** are shown. In FIGS. **33A-33D**, only the second hinge **2760** is shown; however, the first hinge **2756** is identical to or a mirror of the second hinge **2760**, and therefore the description with respect to the second hinge **2760** applies to the first hinge **2756**.

[0171] As shown in FIGS. **33A-33D**, the first hinge element **2764** and the second hinge element **2768** that are pivotably coupled to one another by a hinge pin **2772**. The first hinge element **2764** includes a pawl **3300** that is slidably connected to the first hinge element **2764**. The pawl **3300** includes a first side **3302**, a second side **3304** opposite the first side **3302**, and a first locking surface **3306** extending at least partially from the first side **3302** toward the second side **3304**.

[0172] The second hinge element **2768** includes a hub **3308** that is fixed on position on the second hinge element **2764**. The hub **3308** includes a second locking surface **3310** that can cooperate with the first locking surface **3306** of the pawl **3300** to prevent relative rotational motion between the first hinge element **2764** and the second hinge element **2768** to thereby lock the first hinge **2756** in the first, unfolded, or use position. In some embodiments, for example, as shown in FIGS. **33A-33D**, the first locking surface **3306** and the second locking surface **3310** are flat surfaces that contact one another to prevent relative rotational motion between the first hinge element **2764** and the second hinge element **2768**.

[0173] In some embodiments, the pawl **3300** is moved by a cable **3312**. For example, as shown in FIGS. **33A-33D** and **34A**, a first end of the cable **3312** is connected to the pawl **3300** on the second side **3304** of the pawl **3300**. As shown in FIG. **34A**, for example, the second end of the cable **3312** is connected to a pulley **3314** that is coupled to the handle **114**. The pulley **3314** is configured to rotate when the handle **114** is folded down toward the frame **2716** of the wagon chassis **104**. In some embodiments, the pulley **3314** rotationally fixed to the handle **114** such that the pulley **3314** does not rotate independently of the handle **114**.

[0174] The pawl **3300** is configured to be located in a first, locked position (see FIGS. **33B** and **34A**) and a second, unlocked position (see FIGS. **33D** and **34B**). When the pawl **3300** is in the first, locked position, the first hinge element **2764** and the second hinge element **2768** are unable to pivot with respect to one another. the first hinge **2756** is unable to pivot. When the pawl **3300** is in the second, unlocked position, first hinge **2760** is able to pivot.

[0175] Thus, in some embodiments, to fold the wagon chassis **104**, the handle **114** is rotated toward the frame **2716** of the wagon chassis **104**. Rotation of the handle **114** toward the frame **2716** rotates the pulley **3314**, which in turn pulls on the second end of the cable **3312** connected thereto, wrapping a portion of the cable **3312** around the pulley **3314**. This in turn pulls on the first end of the cable **3312**, which is connected to the pawl **3300**. This pulls the pawl **3300** from its first, locked position (see FIGS. **33B** and **34A**) to its second, unlocked position (see FIGS. **33C**, **33D**, and **34B**). Pulling the pawl **3300** from the first, locked position to the second, unlocked position moves the first locking surface **3306** of the pawl **3300** away from and out of contact with the second locking surface **3310** of the hub **3308**. The hub **3308** and the second hinge element **2768** are then able to rotate with respect to the first hinge element **2764**, allowing the wagon chassis **104** to fold (see FIG. **34B**).

[0176] In some embodiments, the pawl **3300** may be biased toward the first, locked position. For example, with reference again to FIGS. **33B-33D**, the first hinge element **2764** may include one or more biasing elements, such as for example, coil springs **3316** located on the second side **3304** of the pawl **3300** that are configured to exert a force against the pawl **3300** to push the pawl **3300** toward the second hinge element **2768** and into the first, locked position. For example, to unfold the wagon chassis **104**, the handle **114** is rotated away from the frame **2716** of the wagon chassis **104**. Rotation of the handle **114** away the frame **2716** rotates the pulley **3314**, which in turn lets out slack on the second end of the cable **3312** connected thereto, unwrapping a portion of the cable **3312** from around the pulley **3314**. This in turn lets out slack on the first end of the cable **3312**, which is connected to the pawl **3300**. The hub **3308** and the second hinge element **2768** are then rotated with respect to the first hinge element **2764**, allowing the wagon chassis **104** to unfold (see FIG. **34A**). With the slack in the cable, the coil springs **3316** push the pawl **3300** from its second, unlocked position (see FIGS. **33C**, **33D**, and **34B**) to its first, locked position (see FIGS. **33B** and **34A**). Pushing the pawl **3300** from the second, unlocked position to the first, locked position moves the first locking surface **3306** of the pawl **3300** toward and into contact with the second locking surface **3310** of the hub **3308**. The wagon chassis **104** is then locked in the unfolded position.

[0177] Although this embodiment is described as including a cable **3312** connected to the pawl **3300** and the pulley **3314**, it will be understood that in some embodiments, for example, the pawl **3300** may be moved from its first, locked position to its second, unlocked position and/or from its

second, unlocked position to its first, locked position without a cable **3312** and pulley **3314**. For example only, one or more finger pulls may be provided on the frame **2716** of the wagon chassis **104** that a user may be able to manipulate to move the pawl **3300** from its first, locked position to its second, unlocked position and/or from its second, unlocked position to its first, locked position. [0178] Additionally, although FIGS. **33A** and **34A** show the wagon chassis **104** as including two pawls **3300**, it will be understood that in some embodiments, the wagon chassis **104** may be provided with a single pawl **3300** without departing from the scope of the disclosure.

[0179] Referring now to FIGS. **35A-35E**, another embodiment of locking hinges **2756**, **2760** of the wagon chassis **104** are shown. In FIGS. **35A-35E**, only the second hinge **2760** is shown; however, the first hinge **2756** is identical to or a mirror of the second hinge **2760**, and therefore the description with respect to the second hinge **2760** applies to the first hinge **2756**.

[0180] As shown in FIGS. **35A-35E**, the first hinge element **2764** and the second hinge element **2768** that are pivotably coupled to one another by a hinge pin **2772**. As shown in FIGS. **35C** and **35D**, the first hinge element **2764** includes a first plate **3500** and at least one second plate **3502**, the second plate **3502** configured to linearly slide with respect to the first plate **3500**. The second plate **3502** includes two pins, a first pin **3504** and a second pin **3506**. The first hinge element **2764** further includes one or more biasing elements, such as for example, coil spring **3316**.

[0181] The second hinge element **2768** includes at least one hub plate **3508**. In some embodiments, as shown in FIGS. **35C** and **35D**, the second hinge element **2768** includes two hub plates **3508** that are separated a distance to provide a space in which a portion of the first plate **3500** and the second plate **3502** of the first hinge element **2764** are located. That is, a part of the first plate **3500** and the second plate **3502** of the first hinge element **2764** are located between the two hub plates **3508** of the second hinge element **2768**. The hub plates **3508** of the second hinge element **2768** and the first plate **3500** of the first hinge element **2764** are pivotably coupled to one another by the hinge pin **2772**.

[0182] At least one of the hub plates **3508**, and in some embodiments, both hub plates **3508**, includes a first slot **3510** and a second slot **3512**. The second slot **3512** includes a first slot portion **3514** and a second slot portion **3516** connected to the first slot portion **3514**. For example, the first slot portion **3514** is straight and the second slot portion **3516** is arc-shaped having a center that is concentric with the center of the hinge pin **2772**. In some embodiments, the first slot **3510** and the first slot portion **3514** of the second slot **3512** are linearly aligned.

[0183] As shown in FIG. **35B**, when the first hinge **2756** is in the first, unfolded position, the first hinge **2756** may be locked in the first, unfolded position. For example, the first pin **3504** and the second pin **3506** of the second plate **3502** are configured to be located in a first, locked position. In the first, locked position, the first pin **3504** of the second plate **3502** is located in the first slot **3510** of the hub plate **3508** and the second pin **3506** of the second plate **3502** is located in the first slot portion **3514** of the second slot **3512** of the hub plate **3508**. When the first pin **3504** and the second pin **3506** of the second plate **3502** are located in the first slot **3510** and the first slot portion **3514** of the hub plate **3508**, respectively, the first pin **3504** and the second pin **3506** contact the first slot **3510** and the first slot portion **3514** so that the first hinge element **2764** and the second hinge element **2768** are unable to pivot with respect to one another. Stated another way, the cooperation between the first pin **3504** and the second pin **3506** prevent rotation of the first hinge element **2764** with respect to the second hinge element **2768**.

[0184] To fold the first hinge **2756** and the wagon chassis **104**, the second plate **3502** is moved linearly away from the hub plate **3508** such that the first pin **3504** and the second pin **3506** of the second plate **3502** are moved out of the first slot **3510** and the first slot portion **3514** of the hub plate **3508**, respectively. This second, unlocked position is shown in FIG. **35C**. When the second plate **3502** (and the first pin **3504** and the second pin **3506**) is in the second, unlocked position, first hinge **2760** is able to pivot. Stated another way, when the first pin **3504** and the second pin **3506** are no longer received in the first slot **3510** and the first slot portion **3514** of the hub plate **3508**,

respectively, the first hinge element **2764** is able to rotate with respect to the second hinge element **2768**. The wagon chassis **104** may then be folded from the first, use position, as shown FIG. **35A** to the folded, storage position, as shown in FIG. **35E**.

[0185] As described above with respect to FIGS. **33A-33D**, **34A**, and **34B**, the coil spring **3316** located on the second plate **3502** are configured to exert a force against the second plate **3502** to push the first pin **3504** and the second pin **3506** toward and into the first slot **3510** and the first slot portion **3514** of the hub plate **3508**, respectively.

[0186] Although not shown in FIGS. **35A-35E**, the first hinge **2756** and the second hinge **2760** of shown therein may be cable-operated via the handle **114** and/or may be operated, for example only, by one or more finger pulls as described with respect to FIGS. **33A-33D**, **34A**, and **34B**.

[0187] As described herein, in some embodiments, one or more of the first hinge **2756** and the second hinge **2760** are configured to be in a locked position and an unlocked position, wherein in the locked position, one or more of the first hinge **2756** and the second hinge **2760** are configured to retain the wagon chassis **104** in a unfolded configuration, and wherein in the unlocked position, one or more of the first hinge **2756** and the second hinge **2760** are configured to permit the wagon chassis **104** to fold from the unfolded configuration to a folded configuration, and wherein rotating the handle **114** toward the deck **2720** is configured to place one or more of the first hinge **2756** and the second hinge **2760** in the unlocked position.

Frame with T-Slot and Adjustable Ports

[0188] Referring now to FIGS. **36A-36C**, another embodiment of the wagon chassis **104** is shown. In some embodiments, the second frame member **2728** of the frame **2716**, which may in some embodiments be referred to as the rear frame member, includes a T-slot **3600** extending the length of the second frame member **2728**. The T-slot **3600** may extend substantially the width of the wagon chassis **104**. The second frame member **2728** further includes an opening **3602** in the T-slot **3600**. The wagon chassis **104** may be provided with one or more adjustable ports **3604** that can be inserted into the T-slot **3600** through the opening **3602** and can be slid to any desired position along the length of the second frame member **2728** in the T-slot **3600**. As shown in FIGS. **36A-36C**, the adjustable ports **3604** may hang off of the second frame member **2728** toward what may in some embodiments be referred to as the rear of the wagon chassis **104**. For example, each adjustable port **3604** includes a T-slot stud **3606** that may be retained in the T-slot **3600** and an adjustable port body **3608** coupled to the T-slot stud **3606**. In some embodiments, the adjustable port **3604** may include a nut **3610** that may be tightened on the T-slot stud **3606** to secure the adjustable port **3604** in place along the T-slot **3600**. The adjustable port body **3608** includes a socket **3612** that are configured to receive, for example, the pegs **122** of any of the accessories described herein. As shown in FIG. **37B**, a basket **3614** with pegs **122** may be coupled to the wagon chassis **104** by inserting the pegs **122** into the sockets **3612** of the adjustable ports **3604**. As shown in FIG. **37C**, a U-shaped accessory rack **3616** with pegs **122** may be coupled to the wagon chassis **104** by inserting the pegs **122** into the sockets **3612** of the adjustable ports **3604**. The U-shaped accessory rack **3616** may be used to carry various items and/or may be used as a handle to push or pull the wagon chassis **104**.

[0189] Although FIGS. **36A-36C** show the T-slot **3600**, the opening **3602**, and adjustable ports **3604** in and connected to the second frame member **2728** at the rear of the wagon chassis **104**, it will be understood that the T-slot **3600**, the opening **3602**, and adjustable ports **3604** may be provided in and connected to any of the frame members of the frame **2716**, in some of the frame members of the frame **2716**, or all of the frame members of the frame **2716** without departing from the scope of the disclosure.

[0190] The subject matter of embodiments of this patent is described here with specificity to meet statutory requirements, but this description is not necessarily intended to limit the scope of the claims. The claimed subject matter may be embodied in other ways, may include different elements or steps, and may be used in conjunction with other existing or future technologies. This

description should not be interpreted as implying any particular order or arrangement among or between various steps or elements except when the order of individual steps or arrangement of elements is explicitly described.

[0191] Different arrangements of the components depicted in the drawings or described above, as well as components and steps not shown or described, are possible. Similarly, some features and subcombinations are useful and may be employed without reference to other features and subcombinations. Embodiments of the disclosure have been described for illustrative and not restrictive purposes, and alternative embodiments will become apparent to readers of this patent. Accordingly, the present disclosure is not limited to the embodiments described above or depicted in the drawings, and various embodiments and modifications can be made without departing from the scope of the claims below.

[0192] The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the invention, and all such modifications are intended to be included within the scope of the invention.

Claims

1. A wagon assembly, comprising: a modular wagon bed comprising: at least one sidewall positionable in a use position and a stored position; and a footwell feature positionable in an extended position and a storage position; and a wagon chassis comprising: a pivotable handle; a plurality of wheels; and a deck, wherein the modular wagon is removably coupled to the wagon chassis.
2. The wagon assembly of claim 1, wherein the deck comprises a movable surface for accommodating the footwell feature of the modular wagon when the modular wagon bed is coupled to the wagon chassis.
3. The wagon assembly of claim 1, further comprising one or more adapters configured to removably couple to one or more of the wagon chassis and the modular wagon bed and configured to removably couple to an accessory.
4. The wagon assembly of claim 1, wherein the deck includes a population of openings arranged in a plurality of rows and columns.
5. The wagon assembly of claim 4, wherein at least one opening of the population of openings is configured to receive a peg of an additional component to couple the additional component to the wagon chassis.
6. The wagon assembly of claim 1, wherein the handle is configured to transition from a first position and a second position, and wherein the handle includes a pivotable coupling for pivoting the handle between the first position and the second position.
7. The wagon assembly of claim 6, wherein the handle is positionable in a third position via the pivotable coupling for pivoting the handle between the first position and the second position.
8. A foldable wagon assembly, comprising: a wagon chassis comprising: a first chassis portion, comprising a first frame portion and a first deck panel; a handle coupled to the first chassis portion; a first pair of wheels coupled to the first chassis portion; a second chassis portion, comprising a second frame portion and a second deck panel, the second chassis portion connected to the first chassis portion by a first hinge and a second hinge, the first hinge and the second hinge configured to permit the first chassis portion and the second chassis portion to rotate with respect to one another between a first position and a second position; and a second pair of wheels coupled to the second chassis portion.

- 9.** The foldable wagon assembly of claim 8, wherein at least one of the first deck panel and the second deck panel include a population of openings.
- 10.** The foldable wagon assembly of claim 9, wherein at least one opening of the population of openings is configured to receive a peg of an additional component to couple the additional component to the wagon chassis.
- 11.** The foldable wagon assembly of claim 8, wherein the wagon chassis further includes a third deck panel and a fourth deck panel, the third deck panel and the fourth deck panel configured to be in a first position and a second position, wherein in the first position, the third deck panel and the fourth deck panel are flush with the first deck panel and the second deck panel.
- 12.** The foldable wagon assembly of claim 11, wherein in the second position, the third deck panel and the fourth deck panel are spaced a distance away from the first deck panel and the second deck panel to form a footwell.
- 13.** The foldable wagon assembly of claim 11, wherein in the second position, the third deck panel and the fourth deck panel are removed from the wagon chassis.
- 14.** A wagon assembly, comprising: a wagon chassis, comprising: a frame having a first end, a second end opposite the first end, a left side, and a right side; a handle coupled to the frame; a deck coupled to the frame, the deck having a population of openings; and a first pair of wheels coupled to the frame.
- 15.** The wagon assembly of claim 14, wherein at least one opening of the population of openings is configured to receive a peg of an additional component to couple the additional component to the wagon chassis.
- 16.** The wagon assembly of claim 14, wherein the wagon chassis includes a first chassis portion and a second chassis portion, the first chassis portion and the second chassis portion rotatably connected to one another, wherein the wagon chassis is configured to fold and unfold along a fold axis extending from the left side to the right side of the wagon chassis.
- 17.** The wagon assembly of claim 14, further comprising a second pair of wheels coupled to the frame, wherein the second pair of wheels is coupled proximate to the second end of the frame and the second pair of wheels is coupled proximate to the first end of the frame.
- 18.** The wagon assembly of claim 14, further comprising one or more hinges coupled to the frame, the one or more hinges configured to be in a locked position and an unlocked position, wherein in the locked position, the one or more hinges are configured to retain the wagon chassis in a unfolded configuration, and wherein in the unlocked position, the hinges are configured to permit the wagon chassis to fold from the unfolded configuration to a folded configuration, and wherein rotating the handle toward the deck is configured to place the one or more hinges in the unlocked position.
- 19.** The wagon assembly of claim 14, further comprising a second pair of wheels coupled to the frame, wherein the first pair of wheels and the second pair of wheels are removably coupled to the frame, and wherein in a first configuration the first pair of wheels are coupled to the first end of the frame and the second pair of wheels are coupled to the second end of the frame, and wherein in a second configuration the first pair of wheels are coupled to the second end of the frame and the second pair of wheels are coupled to the first end of the frame.
- 20.** The wagon assembly of claim 19, wherein the wheels of the first pair of wheels are configured to swivel with respect to the frame.
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